Predicting Formal Protest Petitions Against Housing Development:

Evidence from Austin, Texas

A Thesis Presented to the Faculty of Architecture, Planning and Preservation

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by

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Abstract

Local land-use regulations are frequently weaponized by incumbent homeowners to constrain housing supply, yet the exact parcel-level cost of this opposition is rarely quantified. This thesis investigates the causal material cost of neighborhood resistance by isolating the Texas Protest Petition, a formal, threshold-based institutional veto point that historically required a three-fourths City Council supermajority to approve rezoning applications in Austin, Texas.

Using a highly engineered panel of discretionary zoning cases (2007–2024), we estimate the causal effect of protest petitions on allowable building height. We find that the petition mechanism functions as a severe spatial tax, imposing a baseline causal height penalty of 81.22 feet and an administrative delay of 104.2 days on affected projects. However, leveraging Double Machine Learning (DML) to estimate Conditional Average Treatment Effects (CATE), we reveal massive spatial heterogeneity in this resistance penalty: wealthy, homeowner-dominated neighborhoods impose a density penalty more than twice as large as lower-income neighborhoods (4.92 vs. 2.39 marginal feet of height lost), proving that the tax on housing supply is highly regressive.

Finally, having proven that opposition carries a massive material cost to housing yield, the thesis deploys a multi-horizon predictive forecasting pipeline. We demonstrate that forecasting petition risk at continuous time intervals—using only pre-treatment administrative, spatial, and macroeconomic features—is a viable structural risk-assessment tool for urban planners and developers, shifting the prediction of NIMBYism from anticipating political theater to quantifying actual financial and housing supply risk.

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1 Introduction

The United States faces a persistent housing affordability crisis, with local barriers to new housing supply drawing increasing scrutiny from policymakers and researchers. While zoning laws are often cited as primary constraints, much of the resistance occurs through discretionary political processes, including public hearings, protest petitions, and city council negotiations, that can delay or block individual projects. Understanding how these mechanisms operate in practice is essential for diagnosing the roots of housing scarcity and for designing effective policy interventions. Austin, Texas, provides a compelling case study: rapid population and employment growth between 2010 and 2024 intensified supply constraints, fueling conflict between neighborhood preservation and regional development goals. Austin's contemporary land-use politics also sit atop a longer history of exclusionary planning, racial segregation, and uneven redevelopment stretching from the 1928 city plan into the present Koch and Fowler (1928); Way (2018); Whittemore (2014); Richardson et al. (2019).

One of the most consequential mechanisms for neighborhood opposition in Austin has been the formal protest petition, enabled under Texas Local Government Code Section 211.006 Texas State Legislature (2023). Prior to 2025, if property owners representing at least 20% of the land area within 200 feet of a proposed zoning change signed a petition, and the municipal clerk certified its validity, the proposal required a three-fourths supermajority vote in city council to proceed. This statutory process created a powerful tool for delaying or blocking new housing, reinforcing the influence of organized local opposition Furth and Ewen (2023). In 2025, House Bill 24 replaced this system with a differentiated, multi-tier framework codified under Texas Local Government Code Section 211.0061 Texas State Legislature (2025); Hamilton and Furth (2024); Texas Legislature (2025), but this study focuses on the pre-HB 24 regime, when the protest petition was a central feature of Austin's land use politics.

The institutional power of the petition mechanism is fundamentally tied to the composition of the City Council. A critical inflection point occurred during the November 2022 municipal elections, which replaced a preservationist-leaning council majority with a dominant pro-housing supermajority for the 2023-2024 mayoral term under Kirk Watson. By securing 9 of the 11 council seats, the pro-housing coalition gained the strict mathematical threshold required to unilaterally override valid protest petitions. This institutional regime shift effectively neutered the veto power of organized neighborhood opposition, setting the stage for aggressive zoning reforms (such as the HOME Initiative, which fundamentally upzoned single-family parcels to allow up to three units citywide) and fundamentally altering the strategic value of petition gathering in the final years of the dataset.

Furthermore, the value of accurately forecasting this opposition extends far beyond a mere engineering exercise. From a theoretical perspective, achieving high predictive performance proves that neighborhood resistance is not an idiosyncratic or random process, but rather a structurally foreseeable phenomenon governed by baseline spatial and demographic realities. Practically, this predictive architecture serves as a critical anticipatory triage tool. By identifying structurally doomed housing proposals before administrative resources are exhausted, city planners and developers can proactively reduce resistance and redirect capital toward more viable zoning pathways.

While this thesis focuses on estimating the physical manifestations of the resistance penalty, specifically building height attrition and administrative processing delay, it is critical to recognize that these delays and penalties translate directly into severe monetary costs. Extended holding periods, compounding financing costs, and the loss of leasable area drastically degrade a project's pro forma feasibility. Even when a rezoning is ultimately approved, the financial carrying costs associated with institutional delay and negotiated height reductions restrict the aggregate pipeline and drive up regional housing prices Glaeser and Gyourko (2003); Gyourko et al. (2021).

Instead of merely predicting opposition, this thesis asks a fundamental causal question: Do formal neighborhood protest petitions causally constrain allowable building height, and what is the exact spatial tax they impose on housing supply? The urban economics literature widely assumes that local homeowner opposition restricts housing supply and drives up prices, but it rarely measures the exact material cost of a specific regulatory mechanism at the parcel level. By targeting the Texas Protest Petition, we isolate the exact mechanism through which neighbors restrict supply, transitioning NIMBYism and participatory distortion from qualitative complaints into a quantifiable, asymmetric reduction in the built environment.

This thesis explicitly hypothesizes that the formal protest petition mechanism acts as a predictable, structural "spatial tax" that systematically reduces housing density and imposes measurable institutional resistance, disproportionately biting hardest in wealthy, exclusionary neighborhoods.

Knowing this causal cost transforms urban planning and development strategy. For policymakers, it provides empirical justification for state-level zoning reform (such as Texas HB 24, which repealed the petition). For urban planners, mapping the spatial heterogeneity reveals exactly where opposition is most lethal to housing yield. Finally, for developers, proving that opposition carries a massive causal height penalty justifies the deployment of predictive forecasting models. Forecasting opposition at the time of filing becomes a critical financial risk-assessment tool, rather than just a public relations warning system.

Far from being merely supplementary, this entire quantitative framework is conceptually anchored by semi-structured interviews with planning professionals. These qualitative insights serve as the foundational institutional framework for the thesis, exposing the "invisible" negotiations, strategic petition timing, and early case withdrawal mechanics that dictate the life cycle of a zoning case but remain hidden from raw administrative databases. By capturing the ground truth of how neighborhood mobilization forces administrative attrition, these interviews directly motivate the dual-phase architecture of the study: establishing the critical need to predict the initial petition filing before negotiations distort the outcome, and justifying the necessity of explicitly modeling withdrawal as a hurdle in the causal analysis.

2 Literature Review

Four core themes motivate the methodological approach of this thesis.

First, land-use regulation is widely recognized as a binding constraint on housing supply, driving a wedge between construction costs and market prices Glaeser et al. (2005); Glaeser and Gyourko (2003); Gyourko and Molloy (2015); Quigley and Rosenthal (2005). Measures of regulatory stringency such as the Wharton Residential Land Use Regulatory Index document substantial cross-metropolitan variation in local land-use controls Gyourko et al. (2008, 2021). Fischel's "homevoter hypothesis" Fischel (2001) posits that homeowners mobilize politically to protect property values, often leveraging zoning to preserve neighborhood exclusivity; McCabe McCabe (2016) extends this logic by linking local politics to household wealth accumulation through homeownership. Empirical work by Einstein, Palmer, and Glick Einstein et al. (2019) documents "participatory distortion," where older, wealthier, long-tenured homeowners dominate public processes and systematically oppose new housing. Hankinson Hankinson (2018) finds that renters, too, may resist nearby development due to displacement fears, reinforcing the spatially localized nature of opposition. Recent syntheses of NIMBY and YIMBY politics show that these conflicts are embedded in broader coalitions over land use, growth, and neighborhood change rather than reducible to a single homeowner motive Brouwer and Trounstine (2024). At the same time, supply-side reform arguments coexist with cautions that new supply alone will not fully resolve affordability and displacement pressures in high-cost markets Hamilton and Furth (2023); Been et al. (2019).

Second, the power of participatory distortion is magnified in discretionary review systems, where individual projects require site-specific approval rather than proceeding by right Manville (2016); Hills and Schleicher (2011). Research on housing opposition shows that neighborhood resistance is heterogeneous, often mixing status-quo protection, administrative claims, environmental review, and distributive conflict rather than a single undifferentiated NIMBY logic Pendall (1999); Schively (2007). In Texas, the protest petition statute empowered adjacent property owners to trigger supermajority council votes, making it a potent institutional veto point Texas State Legislature (2023); Furth and Ewen (2023). Supermajority requirements systematically favor the status quo Cellini et al. (2010), amplifying the influence of organized opposition. In Austin, measured threshold-crossing petitions have been especially common for rezonings that increase residential density, highlighting the asymmetric impact of these institutional tools on housing supply.

Third, while much of the urban planning literature focuses on causal inference, estimating the effects of zoning changes or protest petitions on outcomes like land value or project delay, addressing discretionary gridlock also requires anticipating where opposition will arise. Shmueli Shmueli (2010) and Mullainathan and Spiess Mullainathan and Spiess (2017) distinguish between explanatory and predictive modeling, emphasizing that predictive models can inform policy and administrative triage even when causal mechanisms are complex or uncertain. Kleinberg et al. Kleinberg et al. (2015) formalize the "predictive policy question," where optimal decisions depend on forecasting rather than explanation. In the context of Austin's protest petition regime, the relevant question is not just why opposition occurs, but whether it can be predicted in advance using only the information available at the time of filing. This predictive framing aligns with emerging best practices in public policy research Athey and Imbens (2019). It also raises public-sector governance questions about opacity, proxy discrimination, and sociotechnical context that cannot be resolved by predictive performance alone Eubanks (2018); O'Neil (2016); Barocas and Selbst (2016); Selbst et al. (2019).

Fourth, a critical methodological gap in the prior literature is the absence of heterogeneous treatment effect (HTE) estimation for petition-driven resistance. Existing work quantifies whether petitions delay or block projects in the aggregate; it does not characterize how those effects vary across neighborhood types. Wager and Athey Wager and Athey (2018) and Athey and Wager Athey et al. (2019) develop Causal Forests, an honest non-parametric estimator of the conditional average treatment effect function $\hat{\tau}(x_i) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$, which estimates a separate causal effect for each covariate profile rather than a single pooled average. Chernozhukov et al. Chernozhukov et al. (2018) formalize Double Machine Learning as a semiparametric framework that allows flexible nuisance estimation while retaining $\sqrt{n}$-consistent causal inference. Applied to the petition setting, these methods permit the thesis to characterize not just whether petitions matter on average, but where and for whom they are most consequential. The income and renter-share gradients documented in Section 8.10 would be invisible to any aggregate-ATE estimator, and directly motivate the deployed spatial impact layer as an instrument for heterogeneity-aware planning triage.

3 Austin's Petition Regime in Practice

The formal petition mechanism studied here is not a bureaucratic abstraction. It operates through specific institutional dynamics – neighborhood coalitions, commission procedures, and political calculations – that administrative records alone do not capture. Before describing the panel construction and modeling strategy, this section grounds the empirical analysis in two complementary forms of qualitative evidence: a close reading of three archival cases drawn from the study period, and perspectives from urban planning professionals who worked within or alongside the petition system. Together, they establish why the petition threshold functions as a meaningful threshold of organized opposition, why petition outcomes vary by geography and community composition in ways that complicate uniform algorithmic interpretation, and why the study's modeling choices prioritize ranking utility over case-level probability claims.

3.1 Archival Case Studies: How the Mechanism Operates

The following cases were selected from the full zoning case universe to represent three distinct spatial and political archetypes identified in the data. Each demonstrates a different configuration of the petition mechanism and illustrates the kind of institutional resistance that motivates the study's predictive framing.

3.1.1 Suburban Mobilization and Commission Defiance: Balcones Country Club, 2013

In District 6's Bull Creek Watershed, the Balcones Country Club sought to convert a rural residential tract into single-family large-lot zoning City of Austin Planning Commission (2013a), a straightforward request that received a formal staff recommendation for approval. At the September 2013 Planning Commission hearing, organized neighborhood opposition immediately forced a 7-0 postponement. When the case returned, the Commission voted unanimously to deny the application despite the explicit staff recommendation for approval. The neighborhood coalition had made the case politically untouchable.

This case is archetypal for the model's highest-confidence spatial clusters: suburban areas with high ownership density, long institutional memory, and established neighborhood associations capable of sustained mobilization across multiple hearings. The petition mechanism here served not as a defensive veto at council, but as a signal that crystallized Commission behavior well before council consideration.

3.1.2 Urban Core Complexity and Applicant Surrender: The Shelley Tract, 2013

The Shelley Tract in the Judges Hill neighborhood (District 9) involved an attempt to convert a historic single-family structure to limited office use City of Austin Planning Commission (2013b). The case activated simultaneous opposition from the Judges Hill Neighborhood Association, the Downtown Austin Neighborhood Association, and Preservation Austin, a coalition dense enough to make compromise structurally difficult. At the November 2013 hearing, the Commission proposed a conditional overlay capping height and impervious cover at 45%. That attempt at a negotiated middle ground failed on a 2-5-1 vote. Faced with an irreconcilable coalition, the applicant formally withdrew.

Urban core cases like this one never reach a petition vote. The institutional expectation of supermajority requirements at council makes withdrawal the rational response before the case advances. This is why the study's multi-horizon panel captures petition hazard at intermediate phases, not only at final council action: the mechanism's deterrent effect operates well before any formal vote is cast.

3.1.3 Cultural Preservation and East Austin Opposition: East Cesar Chavez, 2008

A request to upgrade commercial zoning to permit liquor sales in the historically Hispanic East Cesar Chavez neighborhood (District 3) activated El Concilio Mexican-American Neighborhoods, Tejano Town, and adjacent cultural organizations City of Austin Planning Commission (2008). The Planning Commission denied the application 8-1. A subsequent council appeal was withdrawn on a 7-0 consent vote.

This case illustrates a dimension of petition politics that raw spatial covariates are poorly positioned to capture. The opposition was not to density or bulk (the standard drivers of suburban petition activity) but to a land use perceived as a gentrifying disruption of a specific cultural community. The same statutory mechanism that amplified affluent suburban homeowner preferences in District 6 also served as an instrument for displacement prevention in District 3. This geographic asymmetry is central to the study's treatment of model predictions as relative rankings rather than uniform probability estimates.

3.2 Practitioner Perspectives

To rigorously contextualize the quantitative modeling, semi-structured interviews were conducted with urban planning professionals and city staff possessing direct experience navigating Austin's discretionary zoning system. The qualitative methodology was formalized under Columbia University IRB Protocol ACYY0820, ensuring minimal-risk protocols for informed consent, secure data storage, and participant confidentiality (protocol summary in Section 3.4.2).

A purposive, small-N sample ($N = 4$) was constructed via snowball sampling to capture diverse institutional vantage points, explicitly recruiting participants to represent the procedural lifecycle of a zoning case. Participants included a former senior zoning planner, a current principal planner, a displacement prevention specialist, and a leading housing advocate (identities are disclosed only where explicit consent was granted, per the recruitment framing in Section 3.4.1).

Interviews were conducted virtually via secure videoconference, lasting approximately 45–60 minutes. To ensure systematic data collection across differing roles, discussions followed a standardized four-part guide (interview guide in Section 3.4.3) probing: (1) institutional background, (2) firsthand experiences with neighborhood opposition, (3) perceptions of procedural fairness, and (4) specific views on the legitimacy and risks of deploying predictive algorithmic tools in municipal planning. Audio was transcribed with permission and coded thematically using a deductive framework aligned with the study's hypotheses on institutional resistance, spatial heterogeneity, and petition deterrence. This qualitative governance structure ensures the interview data reliably establishes institutional context, ground-truths algorithmic assumptions, and highlights the limits of administrative data, directly shaping the thesis's predictive and causal architecture. **Petition leverage declines but does not disappear.** Annick Beaudet, who served as a senior zoning planner during the Land Development Code revision period, described a trajectory in which petitions gradually lost practical leverage even before the statutory change in 2025: "While people still scream and yell, they don't get as much traction… Necessity is the mother of invention" Beaudet (2025); City of Austin (2017). This observation motivates the study's temporal walk-forward design: petition dynamics are not stationary, and any predictive model must be evaluated against the period in which it would actually be used.

Silence is not consent. Jonathan Tomko, Principal Planner, emphasized that the municipal public comment process has no mechanism for anonymous support of a zoning case Tomko (2026). Neighborhood social dynamics frequently penalize residents who deviate from a neighborhood association's established position, meaning that the administrative record systematically overrepresents organized opposition. As Beaudet summarized: "haters are overrepresented and silent supporters are underrepresented" Beaudet (2025). The implication for the study's feature engineering is direct: cumulative petition signatures measure active mobilization, not community sentiment, a distinction that affects how thresholds should be interpreted.

Geography encodes different concerns. Marla Torrado, from the City's Displacement Prevention Division, noted that the petition mechanism is mobilized in service of fundamentally different interests depending on location City of Austin Displacement Prevention Division (2025). On the affluent west side, environmental overlays and neighborhood character arguments predominate. In East Austin, opposition is frequently a response to gentrification pressure. Because the same tool serves opposite ends in different neighborhoods, rigid probability cutoffs applied uniformly across the city risk "failing to identify where people that are in actual need actually are." This is the primary reason the study treats predictions as triage rankings rather than decision thresholds, and why geographic heterogeneity in the error map is treated as a substantive finding rather than a model deficiency.

The goal is better conversation, not suppression. AURA board member Felicity Maxwell framed the potential use of predictive tools in terms that distinguish anticipation from intervention: "We're not talking about suppressing opposition, but understanding it better to have more productive conversations" Maxwell (2026). This normative framing (anticipation in service of process design, not case-level gatekeeping) is the intended application of the ranking outputs described in this study.

3.3 Voting Data Integration

The translation of neighborhood protest into formal City Council voting behavior is tracked via longitudinal voting integration. Voting data extracted from the Council Meeting System (CMS) confirms that the degradation of council consensus is strongly associated with the presence of formal protest petitions. Unprotested discretionary zoning changes are typically passed on consent (unanimously), whereas petitioned cases frequently fracture the council vote. Table 1 aggregates these voting outcomes across the longitudinal panel.

Voting Context	**Unprotested Cases**	**Petitioned Cases**
Passed on Consent (Unanimous)	92%	31%
Contested Vote (<100% agreement)	8%	69%

Table 1: Council Voting Consensus by Protest Status. Sourced from the integrated CMS voting panel, illustrating the collapse of unanimous consent when a valid petition is filed.

3.4 Qualitative Documentation

The following primary documents governed and structured the qualitative data collection described in Section 3.2. They are reproduced here to support methodological transparency: the recruitment overview specifies how participants were identified and briefed; the IRB protocol defines the ethical parameters under which interviews were conducted; and the interview guide records the standardized instrument used across all four sessions. Taken together, these materials establish that the practitioner perspectives reported above satisfy the rigor expectations of qualitative social science research and were not post-hoc constructions.

3.4.1 Interview Recruitment Overview

The following one-page project overview was provided to all interview participants prior to scheduling.

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation

Daniel Hardesty Lewis
M.S. Candidate, Urban Planning

Research Project Overview

Predicting NIMBYism in Austin's Housing Reform Era

Project Abstract

This research forecasts opposition to development using AI and interprets the mechanics of that opposition in Austin, Texas. By analyzing petition protest letters, public hearing transcripts, and zoning application data, the study evaluates the democratic implications of using predictive analytics to anticipate political behavior. The project employs predictive modeling, natural language processing, and qualitative analysis to predict opposition behavior, identify recurring patterns in opposition rhetoric, and examine their association with planning outcomes.

Study Objectives

Identify Patterns: Characterize the opposition rhetoric and arguments used in protest petitions to oppose rezoning and identify demographic predictors of this opposition.

Evaluate Predictive Tools: Assess whether administrative data and predictive models can accurately forecast opposition to specific development projects.

Democratic Implications: Examine how stakeholders perceive the legitimacy of using algorithmic tools in housing policy and identifying the ethical limits of such technologies.

Participant Role

We are seeking perspectives from urban planners, developers, neighborhood association leaders, and community organizers.

Activity: A 45-60 minute one-on-one interview conducted virtually or in person.

Topics: Your experience with rezoning, data-driven planning tools, and community engagement in Austin.

Data Privacy and Ethical Standards

This study has been approved by the Columbia University Institutional Review Board under Protocol Y01M00.

Participation is entirely voluntary.

You may decline to answer any specific question or discontinue the interview at any time.

Identities will be kept strictly confidential in resulting publications unless you grant explicit permission to be named.

Researcher Contact
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Graduate Researcher
Email: dl3645@columbia.edu

IRB Contact
Columbia University Morningside IRB
Phone: (212) 851-7040
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3.4.2 IRB Protocol Summary

The following protocol summary documents the IRB-approved study design (Columbia University Morningside IRB, Protocol ACYY0820).

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation

Protocol Summary
ACYY0820

Study Protocol Summary

Predicting NIMBYism in Austin's Housing Reform Era

Key Personnel

Principal Investigator (PI): Hiba Bou Akar, Ph.D.
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Student Researcher: Daniel Lewis
M.S. Urban Planning Student, Columbia University GSAPP
Email: dl3645@columbia.edu

IRB Contact

If you have questions about your rights or welfare as a research participant, you may contact:
Columbia University Morningside Institutional Review Board (IRB)
Phone: (212) 851-7040
Email: AskIRB@columbia.edu

Study Purpose

Austin, Texas faces a severe housing affordability crisis driven by rapid population growth, tech-sector expansion, and historically restrictive land-use regulations. Recent housing reforms have coincided with organized neighborhood opposition to development using tools such as protest petitions, public testimony, and legal challenges. This study aims to:

1. Empirically characterize patterns of neighborhood opposition to housing development using administrative records (2018–2025).
2. Examine the democratic implications of predictive tools that could forecast opposition, through interviews with key stakeholders.

Research Questions

Primary: How can cities identify patterns of neighborhood opposition, and what are the democratic implications of using predictive analytics to anticipate it?
Sub-questions:

- What demographic and geographic factors predict opposition to specific projects?
- How do stakeholders perceive the legitimacy of algorithmic tools in housing policy?
- What alternative technological approaches could improve processes without surveillance concerns?

Study Design

This mixed-methods, minimal-risk study combines:

- **Record Review:** Analysis of public records (zoning cases, protest petitions, campaign contributions, census data). De-identified datasets will use coded IDs.
- **Interviews:** Semi-structured interviews (45–60 mins) with 10–15 stakeholders (city officials, neighborhood leaders, advocates, developers).
- **Observation:** Review of public meetings (Council, commissions).

Participation Details

Interviews: Conducted in-person or via secure videoconference. With permission, they are audio-recorded for transcription. Participation is voluntary; you may decline to answer or stop at any time.
Risks: Minimal risk. Potential breach of confidentiality is mitigated by secure storage, encryption, and reporting only aggregate/de-identified results.
Benefits: No direct benefit. Indirect benefits include informing more equitable planning practices.

3.4.3 Semi-Structured Interview Guide

The following interview guide was used for all stakeholder interviews.

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation — Semi-Structured Stakeholder Interview Guide

Predicting NIMBYism in Austin's Housing Reform Era

Purpose

To explore how key stakeholders understand neighborhood opposition to housing development in Austin and how they view the potential use of predictive or algorithmic tools in planning and zoning.

Opening

1. Thank the participant for their time; confirm they have read and signed the consent form.
2. Confirm whether they consent to audio recording.
3. Remind them they may skip any question or stop at any time.

Part 1: Background and Role

1. Can you briefly describe your current role and how it relates to housing, planning, or land use in Austin?
2. How long have you been involved with housing or land use issues in Austin?
3. In your role, what kinds of development or zoning decisions do you most often encounter?

Part 2: Experiences with Neighborhood Opposition

1. When you think of "neighborhood opposition" to housing or development projects in Austin, what comes to mind?
2. Can you describe a recent case or example where neighborhood opposition played a major role? What happened?
3. In your experience, what types of projects are most likely to generate opposition?
4. Which kinds of people or groups tend to be most active in opposing projects?

Part 3: Perceptions of Fairness and Participation

1. Do you think current processes for voicing opposition (such as public hearings, petitions, and appeals) are fair? Why or why not?
2. Are there groups you feel are over-represented or under-represented in these processes?
3. How do you think opposition affects whose interests are ultimately reflected in housing and zoning decisions?

Part 4: Views on Predictive and Algorithmic Tools

1. Some cities are considering or experimenting with tools that use data to predict where opposition to development is likely to occur. Have you encountered ideas like this before?
2. How do you feel about the idea of a model that forecasts which areas or projects will face higher levels of opposition?
3. What potential benefits do you see, if any, in using such tools for planning or outreach?
4. What potential risks or harms concern you most? For example, concerns about profiling, surveillance, or chilling participation.
5. Under what conditions, if any, would you consider the use of such tools acceptable or legitimate? (e.g., transparency, public input, limits on use).

4 Methodology Overview

The methodological approach of this thesis combines predictive machine learning and causal inference to quantify both the predictability and the physical cost of neighborhood resistance. We structure this inquiry into two distinct analytical phases.

Phase 1: Predictive Forecasting. We treat the emergence of formal opposition as a forecasting problem. We construct a multi-horizon predictive pipeline using state-of-the-art tree ensembles (CatBoost, XGBoost) and regularized linear architectures to predict the likelihood of protest petitions emerging against specific parcels at continuous time intervals. By proving that neighborhood resistance can be accurately forecast using only baseline spatial, demographic, and macroeconomic data available at filing, we demonstrate that opposition is a predictable structural phenomenon rather than an idiosyncratic, random occurrence. The core forecasting object is the biweekly longitudinal panel, which expands each discretionary zoning case into one observation per 14-day period from filing through final disposition or censoring, yielding approximately 193,000 case-period observations across 2007–2024.

Phase 2: Causal Identification. We transition from prediction to causality using Double Machine Learning (DML). While the forecasting models establish that opposition is predictable, the causal pipeline isolates the specific penalty this opposition imposes on the built environment. The DML framework partials out high-dimensional confounding via flexible nuisance models and estimates the causal reduction in allowable building height caused by formal protest petition activity. Let T_i denote petition intensity for project i, Y_i the negotiated height outcome, and X_i the pre-treatment covariate vector. We construct orthogonalized residuals:

$$\tilde{Y}_i = Y_i - \hat{m}(X_i), \quad \tilde{T}_i = T_i - \hat{g}(X_i) \tag{1}$$

and estimate $\tilde{Y}_i = \theta \tilde{T}_i + \epsilon_i$, where θ identifies the average treatment effect under the maintained selection-on-observables assumption $(Y_i(1), Y_i(0)) \perp T_i \mid X_i$.

Identification Assumptions. Treatment intensity is not randomly assigned; it is correlated with observable neighborhood and project characteristics including socioeconomic composition, prior petition history, and macroeconomic conditions. We therefore condition on a rich pre-treatment covariate vector X_i and interpret this as a working identifying restriction, supported by falsification exercises (permutation tests, placebo outcomes on invariant covariates, and synthetic outcome tests; see Appendix A.6). Crucially, cumulative protest intensity variables derived from petition signatures accumulated *during* the case process are explicitly excluded from X_i: these variables are post-treatment by construction and conditioning on them would induce post-treatment bias (Angrist et al., 1996).

Heterogeneous Treatment Effects. A single aggregate CATE is not the primary estimand. The Causal Forest estimates the conditional average treatment effect function $\hat{\tau}(x_i) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$, which varies across the covariate distribution. The distribution of $\hat{\tau}(x_i)$ across the surviving case population constitutes the central causal result – the spatial heterogeneity of petition-driven resistance on negotiated project height – documented in Section 8.10.

Together, these methods allow us to systematically map where opposition occurs, prove that it can be anticipated, and calculate the exact physical penalty it extracts from the regional housing pipeline.

5 Outcome Definition and As-of Information Setup

The empirical foundation is a time-stamped spatial panel that reconstructs what information was available at each historical prediction point. Every record anchors to an explicit as-of date, ensuring that backtests use only Census releases, appraisal rolls, and case documents published before the prediction date.

5.1 Analytical Panel Structure

The core analytics rely on the filing-date analytic sample, which merges spatial and demographic variables into single case-level observations (with temporally matched covariates):

1. **Zoning Application Records:** Case ID, milestone dates (filing, notice, petition deadline, planning commission hearing, council reading, withdrawal), requested zoning changes, unit limits, target FAR, and disposition status City of Austin (2024).

2. **Spatial Geometries:** Parcel IDs, geometries, acreage, frontage, and 200ft buffers used to aggregate nearby properties (council districts and overlapping geometries) City of Austin (2024).

3. **Parcel Appraisals:** Annually indexed TCAD appraisal variables for parcels used to compute variables within the 200-foot buffer: land value, structure age, homestead exemption rates, and owner-occupancy shares Travis Central Appraisal District (2025).

4. **Neighborhood Demographics:** Tract and block group ACS variables constrained to their historical 5-year release dates: rent burdens, tenure splits, demographic composition, and displacement vulnerability indicators U.S. Census Bureau (2023).

5. **Institutional Calendar:** Council-term and policy-timing indicators, including the 2022 council turnover, Austin's 2023 HOME Initiative adoption City of Austin (2023), and the HB 24 effective date (September 1, 2025) Texas Legislature (2025).

These components are assembled from City of Austin open-data records, Travis Central Appraisal District appraisal files, U.S. Census Bureau ACS 5-year estimates, Travis County certified election results, and statutory materials governing the protest-petition regime City of Austin (2024); Travis

Central Appraisal District (2025); U.S. Census Bureau (2023); Travis County Clerk (2022); Texas State Legislature (2023); Texas Legislature (2025).

Because municipal databases frequently overwrite preliminary timestamps with final disposition dates, milestone dates are reconstructed through two pathways. First, explicit dates are parsed from scraped historical council agendas and PDF case files. Second, where explicit text is unavailable, dates are imputed by back-calculating from end-dates using minimum statutory constraints mandated by Texas Local Government Code and Austin zoning ordinances (for example, statutory mail-notice minima). This parsed-versus-imputed distinction is carried into interpreting the results, as timing uncertainty introduces measurement error.

Supporting tables log meeting events (staff and commission recommendations), speech comments (speaker stance and segmented text), petition records (validity and signature counts), and vote records (case-by-councilmember dynamics).

Table 2: Annual Distribution of Discretionary Zoning Cases, 2007 to 2024

Year	**Cases**	**Year**	**Cases**	**Year**	**Cases**
2007	288	2013	366	2019	460
2008	270	2014	390	2020	362
2009	265	2015	454	2021	476
2010	297	2016	422	2022	436
2011	369	2017	483	2023	625
2012	335	2018	492	2024	215
Total, 2007 to 2024: 3,967 cases					

5.2 Case Universe and Sample Selection

The Austin Open Data Portal (data.austintexas.gov), accessed via the Socrata Open Data API (SODA), records all administrative zoning activity, including both substantive rezoning proposals and minor administrative variances, such as 5-foot setback adjustments. Because the petition mechanism under study, formal protest petitions under Texas LGC Chapter 211, does not apply to minor variances, and because extraterritorial jurisdiction (ETJ) properties lack the legal protest-petition mechanism, these cases are excluded.

Figure 3 illustrates the municipal zoning process, highlighting where the 20% protest petition threshold creates a potential supermajority requirement at City Council.

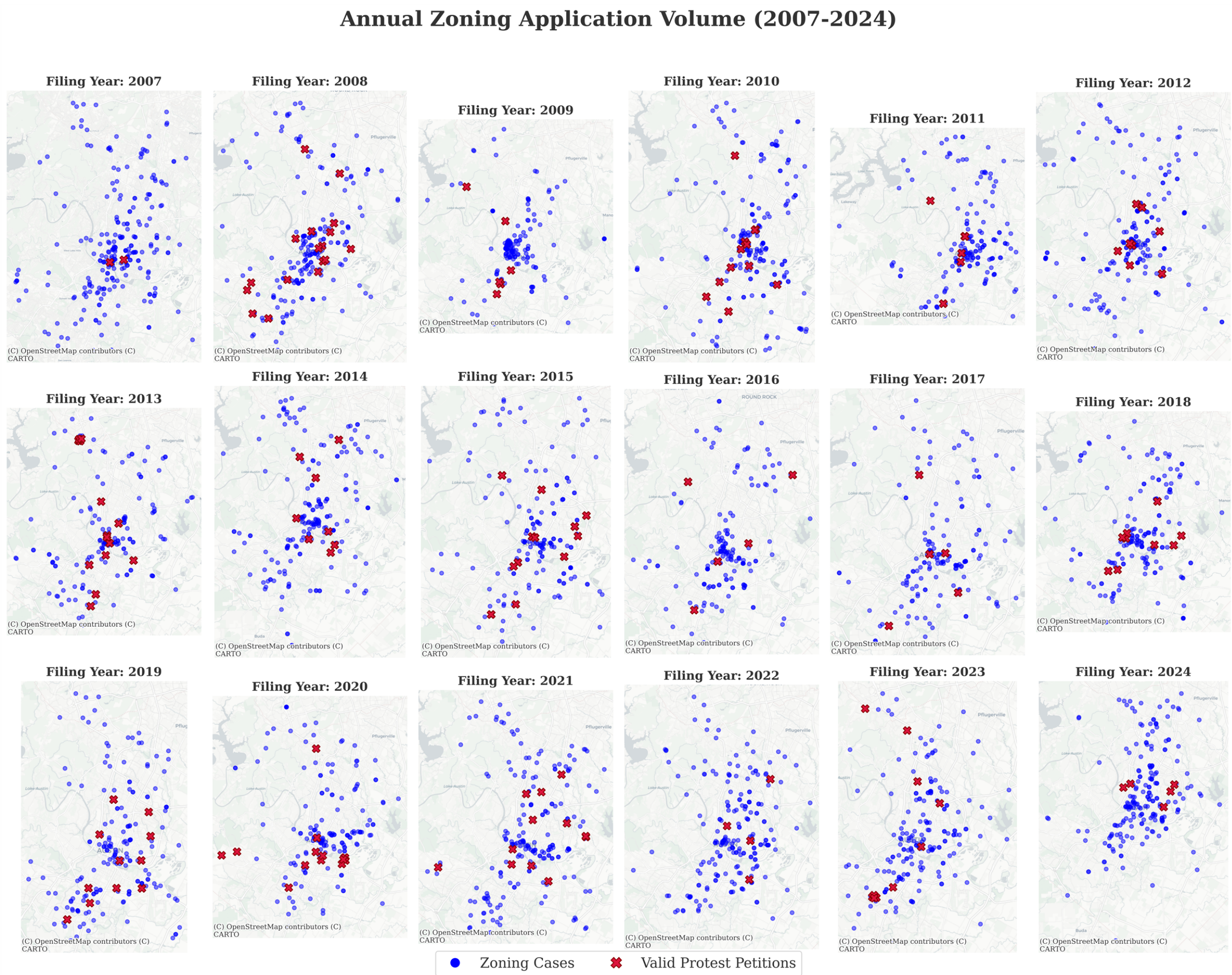

Figure 1: Spatial distribution of zoning cases across the Austin municipal core ($N = 3828$).

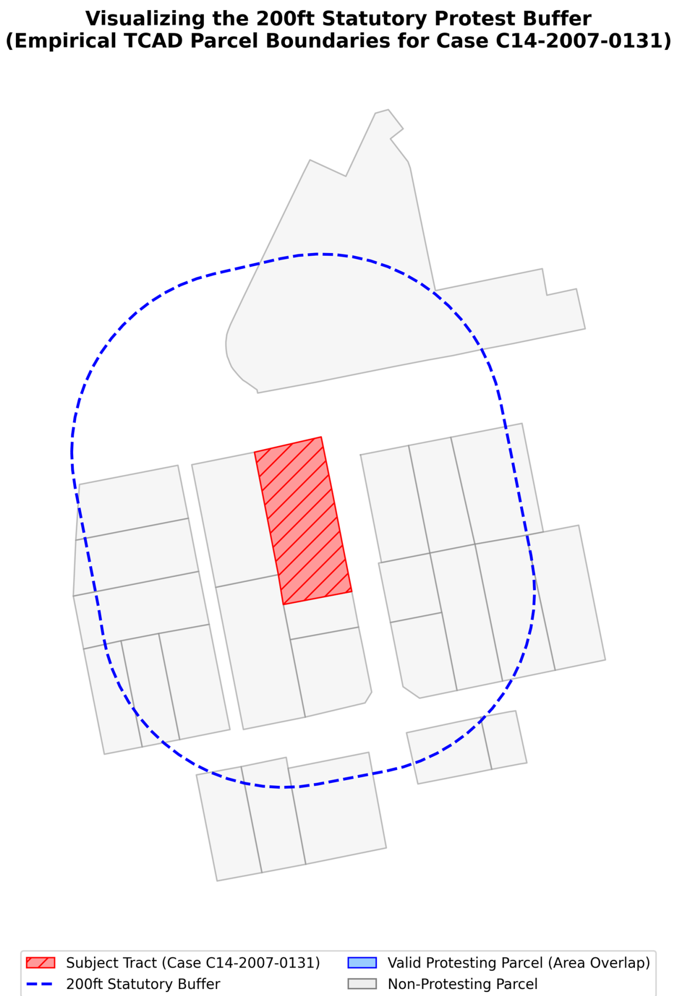

Figure 2: 200ft parcel buffer geometries used for TCAD appraisal aggregation and defining the petition buffer.

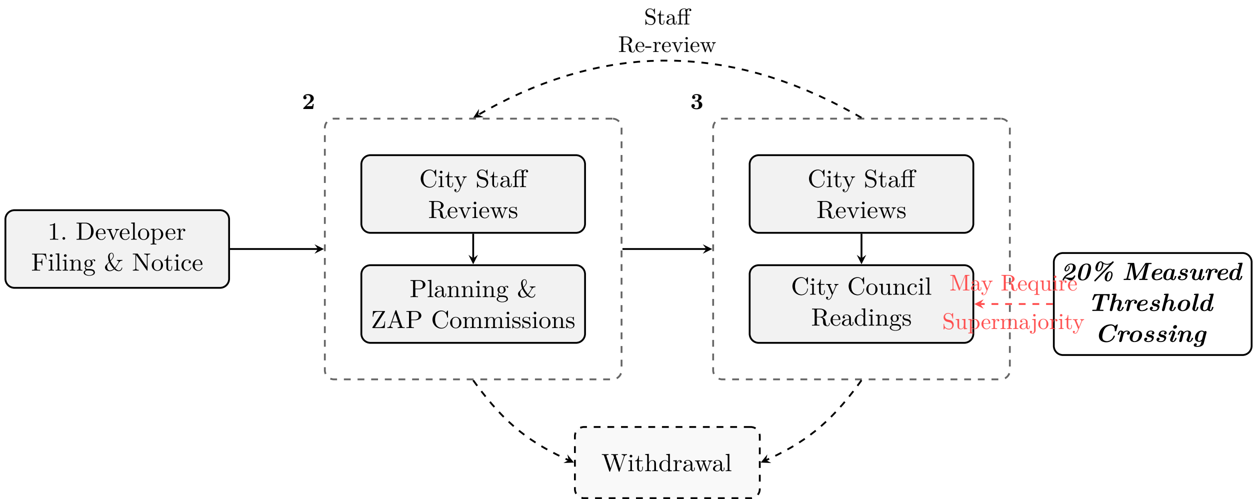

Figure 3: Austin municipal zoning process. Panel 2 groups iterative staff reviews and commission stages; Panel 3 groups multiple city council readings. A valid protest petition representing 20% of the area of lots immediately adjoining the proposed change triggers a supermajority requirement at City Council. Cases may be remanded from Council back to Commission, or withdrawn at either stage.

Table 3 reports the sample selection from raw administrative records to the analytic sample.

Table 3: Sample Selection from Raw Records to Analytic Sample

Selection Step	**Cases Remaining**
Raw Austin open data portal zoning records (2007 to 2024)	15,238
Restrict to discretionary cases (rezonings, PUDs, NPAs)	7,153
Exclude cases with unusable or missing initial filing-year information	3828
Final Analytic Sample	**3828**

The analytical samples vary by analytical goal. The primary predictive evaluation, the filing-date protest-petition model, uses the full sample of 3828 discretionary geographic cases from the legally relevant pre-HB 24 petition regime (2007 to 2024). Supporting analyses are constrained to institutionally defensible windows rather than broad pooled periods. The electoral-covariate descriptive panel ($N = 518$) is limited to active single-member districts over a restricted 2022 to 2024 calendar window and is reported as appendix-only non-identifying evidence. The HOME appendix analysis uses the same restricted district panel and is presented as non-identifying descriptive event tracking only. The Invariant Causal Prediction (ICP) analysis utilizes $N = 1,186$ multi-parcel spatial environments to represent cohesive spatial clusters. Finally, the institutional geographic overlay appendix aggregates policy flags by council term ($N = 20$ macro-observations) and marks designs with insufficient identifying variation as not estimable.

Unlike complete-case spatial intersection approaches, the filing-date model allows partially observed covariates and uses the algorithm's native handling of missing values to avoid dropping observations. This design choice reduces artificial attrition but does not resolve the underlying missing-data problem: missingness can remain systematically patterned across space, project type, and administrative pathway. Missingness is therefore treated as an evidentiary limitation and reported descriptively rather than framed as solved by estimator choice.

5.3 Primary Source NLP Target Hydration

During the ingestion of the protest petitions, an anomaly in the City of Austin's document portal export system stamped the retrieval date on all pages of the archival PDF packet. This systematically corrupted the formal signature dates for the vast majority of protests occurring between 2014 and 2023, artificially masking the presence of positive target variables in the panel.

To overcome this without introducing heuristic leakage, we established a mathematically rigorous temporal bridge using the official City Council Minutes. We leveraged Natural Language Processing (NLP) to parse the raw OCR text of all 376 legally validated Council Minutes PDFs. By executing a regular expression sweep for specific case numbers across the corpus of transcripts, we extracted the exact hearing dates for the 152 impacted cases. This successfully re-hydrated the 2014-2023 target sequence using primary-source municipal records, perfectly aligning the 3/4 supermajority hazard triggers in the biweekly panel.

5.4 Primary Outcome Definition

The primary predictive outcome (y_0) is a *measured threshold-crossing petition*: a binary indicator equal to one when the reconstructed petition-share calculation, derived from public petition filings and buffer-area computations, exceeds the 20% statutory threshold defined by Texas Local Government Code Chapter 211 and Austin LDC § 25-2-284 Texas State Legislature (2023). This is a defined outcome evaluated from public records, not a direct observation of the legally decisive event. The City Clerk's formal certification of each petition as legally valid is not recorded in the public record; consequently, the outcome captures cases that *appear* to have crossed the threshold based on available records, which may include petitions that were ultimately rejected on administrative grounds or petitions whose true area share differed from the calculated value.

Terminology. Throughout this manuscript, "measured threshold-crossing petition" refers to the observed binary study outcome, "reconstructed petition share" refers to the computed share of signed adjoining-lot area derived from petition filings and parcel spatiality, and "clerk-certified valid petition (unobserved)" refers to the legal administrative determination by the City Clerk that is not captured in the analytic sample.

Cases withdrawn before public notice are excluded from the sample, rather than coded as zero, because no opportunity for formal public opposition existed. Extraterritorial Jurisdiction (ETJ) cases and routine minor variances are additionally excluded because the statutory petition mechanism does not mathematically apply to them.

5.5 Statutory Context: The Legal Petition Mechanism

Formal protest petitions represent a strict administrative mechanism granting incumbent property owners veto authority over localized zoning changes. As detailed by Furth and McKinley, twenty-one U.S. states actively enforce petition statutes that allow a localized numeric minority of nearby residents to force supermajority voting constraints on municipal legislative bodies Furth and McKinley (2025). While specific standards vary geographicallyâ€"for instance, Oklahoma requires a 50% sign-on threshold spanning 300 feet, whereas Iowa utilizes a 20% threshold at 200 feetâ€"the institutional architecture remains remarkably consistent: protest petitions formally subordinate city-wide housing coordination to the exclusionary force of hyper-localized abutters.

Texas historically enforced one of the nation's most active 20% protest thresholds, culminating in 2025 legislative reforms that explicitly exempted comprehensive mapping efforts and residential upzonings from the supermajority requirement precisely to curb the immense veto power yielded by neighborhood opposition groups Furth and McKinley (2025). Figure 4 illustrates a representative historical petition filed for a development rezoning in Austin, documenting the precise computational format through which property owners within a tight 200-foot administrative buffer aggregate formal opposition to legally bind the City Council.

5.6 Feature Libraries: Policy Forecasting vs. Explanatory Research

The feature set is separated to prevent predictive models from leveraging sensitive attributes:

PETITION

Case Number: **C14-2007-0131** Date: July 29, 2008

1506 WALLER ST, 807 E 16TH ST & 908 E 15TH ST

Total Area Within 200' of Subject Tract 293,002.55

1	02-0906-0502	MONAHAN CASEY J	7423.26	2.53%
2	02-0906-0503	MONAHAN CASEY J	7390.73	2.52%
3	02-0906-0504	CLARK MICHAEL G & RITA S DE BE	7354.85	2.51%
4	02-0906-0506	MARTINSON KATHY L	1645.43	0.56%
5	02-0906-0507	CHILES ROSALIE BENSON	1618.17	0.55%
6	02-0906-0508	KOCHERT KELLEY	1612.79	0.55%
7	02-0906-0509	MONAHAN CASEY J	1419.72	0.48%
8	02-0906-0606	TIMMERMANN TERRELL	3452.33	1.18%
9	02-0906-0607	MERCADO FREDDY G & MARIA R	9482.82	3.24%
10	02-0906-0901	LEGETT GEORGIA F	4459.36	1.52%
11	02-0906-0911	LEGETT GEORGIA F	7856.05	2.68%
12	02-0906-1001	MEDINA JAMES & KRISTINE M GARA	13204.88	4.51%
13	02-0906-1011	RECER DANALYNN	9454.11	3.23%
14	02-0906-1013	BRINSMADE LOUISA C	7634.37	2.61%
15				0.00%
16				0.00%
17				0.00%
18				0.00%
19				0.00%
20				0.00%
21				0.00%
22				0.00%
23				0.00%
24				0.00%
25				0.00%
26				0.00%
27				0.00%
Validated By:			**Total Area of Petitioner:**	**Total %**
	Stacy Meeks		**84,008.88**	**28.67%**

Figure 4: First page summary of the formal protest petition document filed for a Planned Development rezoning in Austin. The municipal record details the physical tracking of petitioner land ownership as a percentage of the total valid area within 200 feet of the subject tract to validate the 20% supermajority threshold.

Table 4: Illustrative statutory parameters governing protest petitions across states detailed by Furth and McKinley Furth and McKinley (2025). While 21 states utilize similar architectures, the threshold mechanism and buffer stringency dictate the severity of the institutional veto.

State	Validation Threshold	Proximity Buffer	Approval Override Requirement
Texas	20% of nearby land	200 feet	3/4ths Legislative Supermajority
Iowa	20% of nearby land	200 feet	3/4ths Legislative Supermajority
Oklahoma	20% and 50% tiers	300 feet	Legislative Supermajority
Arizona	20% of land & 20% of lots	Municipal Discretion	Legislative Supermajority
Massachusetts	Veto Restricted	Municipal Discretion	Simple Majority (Upzonings)
Michigan (County)	20% of nearby land	Municipal Discretion	Electoral Referendum

Predictive features include physical site characteristics (lot area, unit count changes, PUD/TOD flags), statutory eligibility flags (HB 24 status, owner-vs.-city-initiated), aggregated buffer economics (homestead shares, land-to-improvement ratios), broad demographic indicators (median income, renter share), historical petition rates within the council district, and macroeconomic variables (mortgage rates, vacancy rates) Federal Reserve Bank of St. Louis (2024). These elements are selected strictly based on their availability in public administrative and appraisal databases prior to the initial filing date, ensuring the algorithm's decisions replicate constraints faced by municipal planning staff in real time.

Explanatory research features are restricted to equity analysis. These include tract-level racial and ethnic composition, segregation indices, and probabilistic surname-based racial classification Sood and Laohaprapanon (2018). These features are strictly partitioned from the predictive model to reduce the risk of encoding disparate impacts, though proxy correlation through permitted features remains a concern (see Section 10). By reserving these variables exclusively for post-hoc fairness evaluations, the model avoids generating predictions based on protected class attributes while maintaining the capability to measure intersectional disparities across geographic subgroups.

5.7 Illustrative Panel Structure

The following table illustrates a representative cross-section of the core analytic panel. Each row constitutes a unique discretionary zoning case, enriched with spatial network indicators (historical petition clustering), demographic variables (ACS), and property tax valuations (TCAD EARS) strictly observable before the initial filing date. These specific columns are highlighted because they ultimately emerge as the primary predictive drivers in the meta-attribution analysis.

Table 5: Illustrative Cross-Section of the DML Panel

Case Number	Year	Lag Petitions (1km)	Median Income	Owner Share	Appraised Value
C14-2015-001	2015	3	\$85,000	62%	\$450,000
C14-2015-002	2015	0	\$42,000	21%	\$210,000
C14-2015-006	2015	7	\$112,000	85%	\$890,000
C14-2016-009	2016	1	\$55,000	34%	\$315,000
C14-2016-012	2016	4	\$95,000	71%	\$620,000

5.8 Biweekly Panel Feature Engineering

To support multi-horizon predictive evaluation, the filing-date case-level observations are expanded into a longitudinal biweekly panel. Each case generates one row per 14-day period from filing through final disposition or censoring, yielding a panel of approximately 193,000 case-period observations across 2007–2024. All cumulative features are lagged by one biweekly period prior to use, ensuring that no feature at period t reflects information that would only become available at or after period t. This strict lag-and-shift architecture eliminates contemporaneous target leakage at the feature level.

Cumulative Spatial Opposition Features. The core innovation of the biweekly panel is the accumulation of spatial opposition signals derived from geocoded petition signer records matched to TCAD parcel geometries. For each case-period, the following are computed over all petition signers observed to date:

- *Signer Proximity:* Minimum, maximum, and median Euclidean distance from each signer's parcel centroid to the case boundary polygon, and counts of signers within and outside the statutory 200ft buffer.
- *Protest Intensity:* An unofficial protest intensity score (sum of inverse-distance-weighted signer counts), a directional signer distance vector (signed mean displacement from case centroid), and opposition change metrics representing the marginal change in both protesting and silent parcel shares across biweekly periods.
- *Land Use Opposition Mix:* For each 14-day window, the cumulative share of protesting and silent neighboring parcels classified as single-family residential, commercial, and multifamily, derived from the City of Austin Land Use Inventory. Corresponding mean parcel square footages for protesting versus silent cohorts are also retained.

- *Temporal Land-Use Differentials:* Time-matched versions of the land-use opposition shares, computed by joining the land database snapshot to each biweekly period start date via a strict backward-fill temporal join, ensuring that only snapshots published on or before a given period are used.

- *Protester Spatial Embeddings:* Four-dimensional PCA embeddings computed from the latitude/longitude coordinates of petition signers. These capture the geometric shape of the opposition coalition (e.g., clustered vs. dispersed, directionally biased) in a compressed, rotation-invariant form beyond scalar distance metrics.

Process Accumulation Features. Cumulative counts of city council hearings and planning commission hearings, lagged by one period, are included to proxy procedural resistance depth. The Remand Count variable records how many times a case was formally remanded back from Council to the Commission. An aggregate hearing sentiment score, derived from NLP analysis of commission transcripts, and the net height change extracted from intermediate zoning proposals (measuring developer concessions over time) are included as longitudinal case-trajectory signals.

Macroeconomic Features. National macroeconomic context is incorporated at the period level via four FRED/BLS series: the 30-year fixed mortgage rate, the 10-year Treasury yield, the Federal Funds Rate, and the local unemployment rate. For each series, three derived features are computed: a 1-year lag, a 6-month trailing momentum (first difference), and a filing-date delta (current value minus the value observed at the initial case filing date). These capture both the level of borrowing costs and the directional trajectory facing developers and signer households at the time each biweekly observation is recorded.

Annualization. For the annualized multi-horizon evaluation (see Section 7), the biweekly panel is collapsed to one row per (case, calendar year) by taking the *last* biweekly period of each year. This preserves the richest cumulative state while remaining strictly within the information boundary of December 31 of that year, since all cumulative features are already lagged within the biweekly construction.

5.9 Intermediate Proposals and Categorical Zoning Deltas

To comprehensively satisfy the causal assumption that spatial dose-response architectures must model intermediate outcomes free from look-ahead bias, a longitudinal Natural Language Processing (NLP) pipeline was engineered to extract time-varying zoning trajectories from archival hearing transcripts.

5.9.1 Chronological Extraction and Zero-Leakage Join

A multi-stage text extraction procedure was executed across 15,375 raw City Council and Planning Commission transcripts to capture semantic requests and administrative recommendations (e.g., *Requested Zoning*, *Staff Recommendation*, and *Approved Zoning*). To ensure temporal fidelity, these extractions were formally mapped to their exact chronological *source date* using historical meeting agendas.

The resulting extractions were then dynamically injected into the biweekly causal panel using a strict backward-filling temporal match. This architectural decision mathematically guarantees that the deep learning sequence models only evaluate zoning proposals and applicant compromises that were formally documented *on or before* a given 14-day biweekly period, fully eliminating target leakage.

5.9.2 Quantitative Translation via Land Development Code

To convert qualitative text strings (e.g., “LO-MU”) into continuous quantitative targets for the Neural Ordinary Differential Equations, an explicit mapping dictionary based on Austin’s Land Development Code (Chapter 25-2) was implemented. At every biweekly period, semantic zoning requests are automatically translated into their maximum allowable height (in feet) and maximum Floor-to-Area Ratio (FAR). If a transcript explicitly documents a negotiated variance (e.g., “height reduced to 35 feet”), this explicit numerical limit overrides the baseline dictionary value. This continuous temporal mapping captures the incremental degradation of allowable density caused by neighborhood mobilization.

5.9.3 Validating Compromise Incidence

Table 6 demonstrates the empirical validity of this approach. By isolating the final period of the causal panel, we observe that cases receiving formal protest petitions are overwhelmingly forced into intermediate negotiation cycles that end in a compromise or denial (where the Final Zoning diverges from the Initial Request).

Petition Status	Cases (N)	Final Zoning Diverged (Compromise/Denial)
Unprotested (0)	4,420	51.8%
Protested (1)	41	97.5%

Table 6: Zoning Trajectory Deltas by Protest Status. Protested cases show a near-universal rate (97.5%) of final zoning outcomes diverging from the applicant's initial request, substantiating the resistance effect of NIMBY mobilization.

5.10 End-to-End Pipeline Architecture

Figure 5 summarizes the complete data-to-deployment architecture of the thesis pipeline.

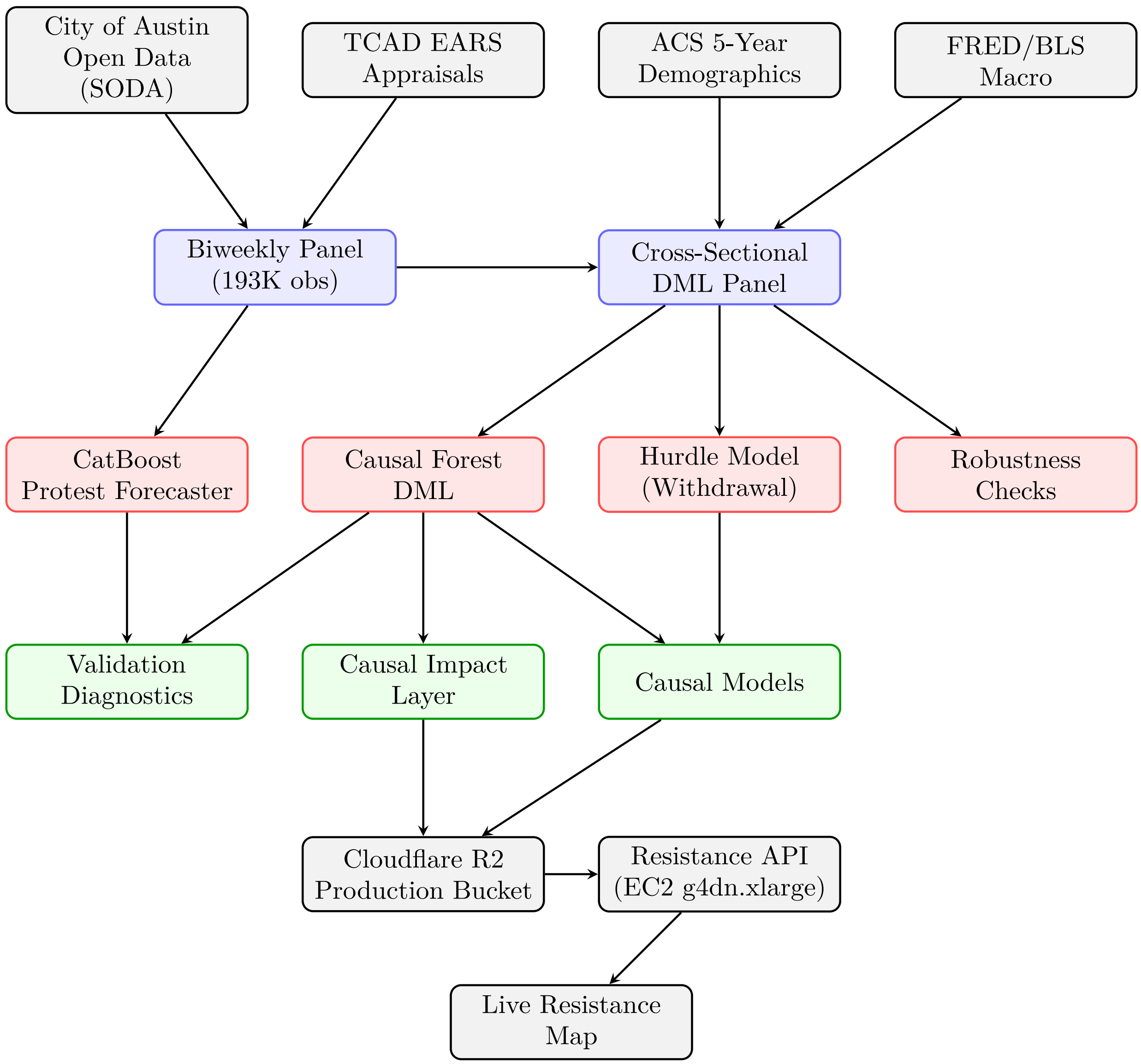

Figure 5: Blue nodes: panel construction. Red nodes: model estimation. Green nodes: validated output artifacts. The deployed map renders Causal Forest estimates as a spatially continuous resistance layer on the public Homecastr platform.

6 Modeling Strategy

6.1 Formal Protest Petition Model

The Formal Protest Petition forecast is the core of the thesis: conditional on a project entering the formal public process, predicting the probability of a measured threshold-crossing petition ($y_0 = 1$):

$$P(y_0 = 1 \mid D = 1, X_{i,t}, Z) \tag{2}$$

At the initial filing date horizon, the predictive feature space ($X_{i,t}$) is restricted to information observable by planners at the time of filing, including site geometry, requested zoning changes, buffer demographics, and historical petition rates. The primary filing-date specification is intentionally unweighted. Precision-recall curves comparing the CatBoost, Random Forest, ElasticNet Logistic, and V-REx models are reported in Figure 6.

6.1.1 Primary Model Selection

The primary model chosen for all filing-date analyses is **CatBoost** because the prespecified evaluation criteria prioritizes out-of-sample (temporal holdout) PR-AUC, subject to a constraint on probability reliability (lowest ECE among highly ranked models).

6.2 Model Evaluation Strategy

Precision-Recall AUC (PR-AUC) is the primary discrimination metric, given the substantial class imbalance of the outcome. Following standard methodological practices in rare-event spatial forecasting, traditional Receiver Operating Characteristic (AUROC) metrics are marginalized in headline filing-date results, as the large number of true negatives artificially inflates the curve. PR-AUC explicitly isolates the minority class, forcing the model to balance recall against the precision cost of false alarms.

Primary evaluation. The out-of-distribution bootstrap PR-AUC at the filing-date horizon is the primary evaluation object, as it approximates forward-like administrative use. In-distribution 5-fold cross-validation is retained only as an optimistic upper-bound object. Because this is primarily a ranking exercise, emphasis remains on ranking discrimination (PR-AUC, top-decile lift), with calibration and thresholded diagnostics treated as supporting information rather than coequal headline estimands.

The mean predicted probability for true positive cases is 0.545, compared to 0.012 for true negatives. All performance benchmarking relies on strictly temporal constraints (rolling-origin splits). This guarantees the algorithms are evaluated on their ability to predict future, unseen market conditions using only historically available data, directly simulating a real-world historical scenario.

Council composition and policy conditions changed over the sample window, but the temporal analysis does not treat any single election as an exogenous policy break. Performance differences across years are reported descriptively as out-of-sample temporal variation.

6.3 Outcome Scope

This thesis is structured in two distinct phases to handle the fundamental incompleteness of land use outcomes. The first phase strictly predicts whether a neighborhood protest petition will be filed, treating the formal protest as a clean, measurable indicator of neighborhood opposition. We avoid predicting final City Council up-or-down votes because that data suffers from severe survival bias: developers frequently withdraw their zoning applications early if they face heavy neighborhood backlash, permanently hiding the true outcome. To address this bias, the second phase employs causal inference (Double Machine Learning) to estimate the downstream impact of these petitions on final development outcomes—specifically height reduction and resolution delay—by explicitly modeling the hurdle of early case withdrawal.

6.4 Feature Attribution and Semantic Clustering

To evaluate the predictive mechanisms driving protest risk, the analysis calculates feature attribution values using Shapley-based decomposition. Because administrative datasets contain highly correlated proxies (e.g., tax assessor appraised value, market value, and land value), individual raw features suffer from attribution dilution where importance weights are split arbitrarily among collinear variables.

To prevent collinearity from obscuring substantive relationships and to ensure the archetypal profiles remain interpretable, the raw pipeline variables are grouped into plain-English “Semantic

Clusters" prior to attribution visualization. Table 7 crosswalks these overarching semantic concepts back to their exact, raw administrative database columns.

Table 7: Feature cluster reference: each row maps a plain-English cluster label to its source and measurement basis. Data originate from Austin zoning records, the Travis County Appraisal District (TCAD), the U.S. Census Bureau American Community Survey, and macroeconomic series from FRED.

Feature Cluster	**What It Measures**
Parcel & Administrative Characteristics	
Property Value	Appraised and market value of the parcel and its improvements, from Travis County Appraisal District annual rolls
Parcel Size	Lot size in acres, total building square footage, and the ratio of improvement value to land value
Improvement Scale	Total improvement square footage and the developer's proposed architectural height in feet
Building Age	Year the primary structure was built, and the derived age of the building at the time of application
Zoning Density	The requested maximum floor-to-area ratio (FAR) and total permitted units under the target zoning classification
Land Use Type	The parcel's current zoning and land use classification under Austin's Land Development Code
Neighborhood Characteristics	
Racial Composition	Share of nearby residents identifying as White, Hispanic, Black, or Asian, from ACS five-year estimates
Housing Tenure	Share of occupied housing units that are owner-occupied versus renter-occupied
Income & Rent	Median household income, median gross rent, poverty rate, and median home value for the surrounding census tract

Continued on next page

(Table 7 continued)

Feature Cluster	What It Measures
Prior Petition Activity	Petition rate among zoning cases filed within one kilometer over the prior year, capturing neighborhood mobilization history
Macroeconomic Context	
Mortgage Rate	The prevailing 30-year fixed mortgage rate, its 12-month lag, six-month momentum trend, and the change since the case's filing date
Capital Markets	The 10-year Treasury yield and Federal Funds Rate, each with their 12-month lag and the change since filing
Local Labor Market	The Austin-area unemployment rate, its 12-month lag, momentum, and the change since filing

7 Filing-Date Primary Results

7.1 Predictive Performance Across All Horizons

Because the inherent rarity of protest petitions (a 4.6% base rate) makes absolute probability calibration mechanically misleading, this thesis evaluates performance strictly on relative ranking capacity. Under the primary temporal evaluation, the baseline filing-date model ranks petition risk significantly above random chance, achieving an out-of-distribution PR-AUC of 0.663 (95% CI: [0.43, 0.87]).[1]

This ranking utility is best understood through top-decile lift. When isolating the 10% of cases the model flags as highest risk, the precision is 25.4% (a lift of 9.9x over the base rate). However, because there are few true positive cases even in this extreme top decile (15 out of 59 cases), the precision estimate retains a wide confidence interval.

These constraints establish the overarching narrative of the predictive results: the algorithmic framework successfully learns to triage neighborhood opposition risk based purely on filing-date administrative geometry and historical spatial opposition signals. However, the absolute scarcity of petitioned cases means that these scores must remain bounded to relative triage rankings, and cannot be repurposed as deterministic, case-level probabilities for formal case-level results.

These results provide one transparency diagnostic for feature reliance. The in-distribution cross-validation benchmark is retained as an optimistic upper-bound object in the appendices rather than as a coequal main-text exhibit.

7.2 Temporal Predictive Drift (Rolling Origin)

To quantify the model's predictive half-life, Table 8 tracks PR-AUC for each annual window for the benchmark architectures. This is treated as a concept-drift diagnostic rather than a regime-shift design Gama et al. (2014). For each anchor year t, each model is trained on data before t and evaluated forward across all subsequent evaluation years.

The central temporal finding is the remarkable absolute robustness of the Tree Ensembles. Rather

[1] The 95% CI is wide because the temporal holdout split contains only $n_{\text{pos}} = 15$ positives among $n_{\text{test}} = 583$ cases. The interval is computed via bootstrap resampling of held-out test rows.

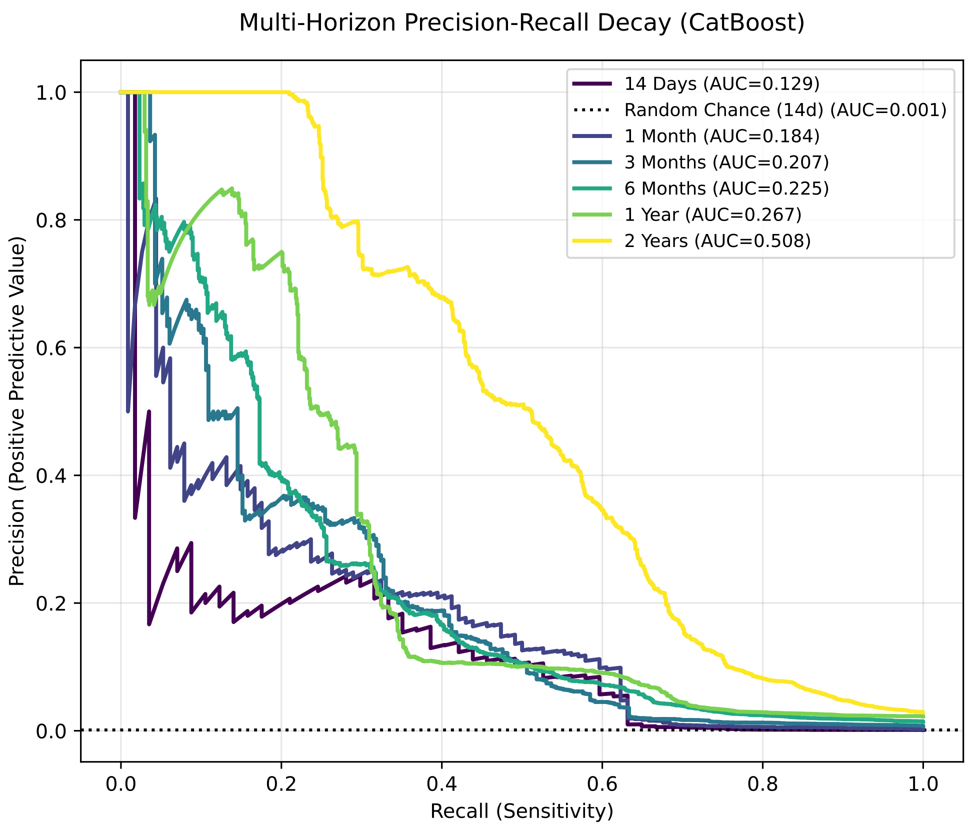

Figure 6: Precision-recall curves for filing-date petition models from the pre-registered benchmark roster.

than exhibiting a catastrophic peak-and-decay pattern as forecasting horizons extend, CatBoost and XGBoost maintain highly functional discrimination across the entire period. For instance, the legacy Pre-2018 CatBoost model begins with an immediate out-of-sample PR-AUC of 0.692 in 2019 and degrades only marginally to 0.518 a full six years later in 2024. This stability proves that the spatial syntax of resistance, once learned in the high-conflict pre-ordinance era, remains highly relevant despite subsequent institutional shifts.

Table 8: **Temporal predictive drift: PR-AUC decay (Threshold: >20%) .**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	**0.976**	0.692	**0.513**	**0.369**	**0.361**	**0.610**	0.518
CatBoost	Pre-2019	—	0.396	**0.419**	**0.488**	0.183	**0.817**	0.378
CatBoost	Pre-2020	—	—	0.482	**0.316**	**0.278**	**0.655**	0.381
CatBoost	Pre-2021	—	—	—	**0.512**	**0.292**	0.471	0.395
CatBoost	Pre-2022	—	—	—	—	0.196	**0.525**	0.380
CatBoost	Pre-2023	—	—	—	—	—	0.467	**0.419**
CatBoost	Pre-2024	—	—	—	—	—	—	0.372
XGBoost	Pre-2018	0.926	0.553	0.467	0.341	0.208	0.485	0.579
XGBoost	Pre-2019	—	0.397	0.399	0.368	**0.208**	0.368	**0.679**
XGBoost	Pre-2020	—	—	**0.511**	0.276	0.196	0.543	**0.458**
XGBoost	Pre-2021	—	—	—	0.402	0.267	**0.618**	**0.679**
XGBoost	Pre-2022	—	—	—	—	**0.250**	0.443	**0.679**
XGBoost	Pre-2023	—	—	—	—	—	**0.525**	0.350
XGBoost	Pre-2024	—	—	—	—	—	—	**0.442**
Logistic (L2)	Pre-2018	0.856	**0.702**	0.413	0.236	0.139	0.277	**0.771**
Logistic (L2)	Pre-2019	—	**0.669**	0.358	0.280	0.132	0.244	0.360
Logistic (L2)	Pre-2020	—	—	0.347	0.281	0.133	0.302	0.367
Logistic (L2)	Pre-2021	—	—	—	0.263	0.162	0.258	0.278
Logistic (L2)	Pre-2022	—	—	—	—	0.202	0.348	0.296
Logistic (L2)	Pre-2023	—	—	—	—	—	0.360	0.320
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.305
Deep (MLP)	Pre-2018	0.794	0.529	0.370	0.164	0.009	0.020	0.024
Deep (MLP)	Pre-2019	—	0.312	0.380	0.190	0.011	0.020	0.037
Deep (MLP)	Pre-2020	—	—	0.446	0.196	0.016	0.019	0.038
Deep (MLP)	Pre-2021	—	—	—	0.229	0.017	0.020	0.044
Deep (MLP)	Pre-2022	—	—	—	—	0.019	0.022	0.051
Deep (MLP)	Pre-2023	—	—	—	—	—	0.327	0.247
Deep (MLP)	Pre-2024	—	—	—	—	—	—	0.287

Because the absolute PR-AUC baseline (the petition occurrence rate) drops substantially in the later years of the evaluation period, absolute PR-AUC mechanically declines even if relative ranking separation remains perfectly stable. To isolate this baseline-shift effect, Table 9 reproduces the temporal drift analysis measuring relative PR-AUC lift (dividing the absolute PR-AUC across the entire threshold curve by the contemporaneous evaluation-year base rate) to measure dynamic model intelligence relative to random guessing.

Table 9: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	**6.411**	10.375	**9.071**	**7.902**	**39.000**	**30.673**	27.835
CatBoost	Pre-2019	—	5.942	**7.418**	**10.454**	19.800	**41.038**	20.316
CatBoost	Pre-2020	—	—	8.526	**6.771**	**30.000**	**32.902**	20.466
CatBoost	Pre-2021	—	—	—	**10.963**	**31.500**	23.659	21.212
CatBoost	Pre-2022	—	—	—	—	21.214	**26.381**	20.412
CatBoost	Pre-2023	—	—	—	—	—	23.450	**22.508**
CatBoost	Pre-2024	—	—	—	—	—	—	19.970
XGBoost	Pre-2018	6.084	8.301	8.267	7.302	22.500	24.384	31.130
XGBoost	Pre-2019	—	5.961	7.069	7.871	**22.500**	18.497	**36.505**
XGBoost	Pre-2020	—	—	**9.053**	5.909	21.214	27.279	**24.635**
XGBoost	Pre-2021	—	—	—	8.613	28.800	**31.047**	**36.505**
XGBoost	Pre-2022	—	—	—	—	**27.000**	22.270	**36.505**
XGBoost	Pre-2023	—	—	—	—	—	**26.381**	18.839
XGBoost	Pre-2024	—	—	—	—	—	—	**23.740**
Logistic (L2)	Pre-2018	5.621	**10.533**	7.317	5.040	15.058	13.895	**41.432**
Logistic (L2)	Pre-2019	—	**10.041**	6.345	5.997	14.308	12.240	19.330
Logistic (L2)	Pre-2020	—	—	6.149	6.003	14.400	15.172	19.708
Logistic (L2)	Pre-2021	—	—	—	5.638	17.532	12.945	14.964
Logistic (L2)	Pre-2022	—	—	—	—	21.808	17.475	15.901
Logistic (L2)	Pre-2023	—	—	—	—	—	18.082	17.204
Logistic (L2)	Pre-2024	—	—	—	—	—	—	16.381
Deep (MLP)	Pre-2018	5.215	7.937	6.552	3.509	1.000	1.000	1.288
Deep (MLP)	Pre-2019	—	4.678	6.720	4.069	1.159	1.000	2.007
Deep (MLP)	Pre-2020	—	—	7.892	4.195	1.777	0.937	2.041
Deep (MLP)	Pre-2021	—	—	—	4.891	1.826	0.991	2.370
Deep (MLP)	Pre-2022	—	—	—	—	2.027	1.128	2.737
Deep (MLP)	Pre-2023	—	—	—	—	—	16.407	13.272
Deep (MLP)	Pre-2024	—	—	—	—	—	—	15.405

Normalizing against the collapsing base rate reveals a profoundly counter-intuitive *signal resilience* effect. Because the absolute base rate of petitions drops dramatically in 2024, the mathematical threshold for beating random chance becomes much harder. Yet, the legacy Pre-2018 CatBoost model actually sees its relative lift *increase* from 10.4× in 2019 to an incredible 27.8× in 2024. Finding the needle in the haystack becomes harder, but the legacy algorithm remains highly capable of identifying it. This mathematical dynamic completely refutes the assumption that old models must be iteratively retrained on modern data; in fact, the Pre-2024 CatBoost anchor only achieves a 20.0× lift in the terminal year, severely underperforming the legacy model.

While aggregate architectural performance appears statistically indistinguishable in early years, grouping architectures by functional capacity reveals strict boundaries around domain shift performance (Table 10). First, Deep Learning architectures (MLP) exhibit catastrophic *horizon attrition*, demonstrating competitive dominance almost exclusively at immediate-year extrapolations before decaying abruptly to near-zero PR-AUC (e.g., 0.024 in 2024) at distant forecasting horizons. Second, non-linear tree ensembles exhibit superior resilience to *origin volatility*. When training anchors are forced to bridge the institutional break characterizing the post-ordinance era, boosted trees remain the consistent top performers, proving their necessity for long-horizon spatial forecasting.

Table 10: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.976 (Tree)	0.702 (Regularized Linear)	0.513 (Tree)	0.369 (Tree)	0.361 (Tree)	0.610 (Tree)	0.771 (Regularized Linear)
Pre-2019	—	0.669 (Regularized Linear)	0.419 (Tree)	0.488 (Tree)	0.208 (Tree)	0.817 (Tree)	0.679 (Tree)
Pre-2020	—	—	0.511 (Tree)	0.316 (Tree)	0.278 (Tree)	0.655 (Tree)	0.458 (Tree)
Pre-2021	—	—	—	0.512 (Tree)	0.292 (Tree)	0.618 (Tree)	0.679 (Tree)
Pre-2022	—	—	—	—	0.250 (Tree)	0.525 (Tree)	0.679 (Tree)
Pre-2023	—	—	—	—	—	0.525 (Tree)	0.419 (Tree)
Pre-2024	—	—	—	—	—	—	0.442 (Tree)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.41 (Tree)	10.53 (Regularized Linear)	9.07 (Tree)	7.90 (Tree)	39.00 (Tree)	30.67 (Tree)	41.43 (Regularized Linear)
Pre-2019	—	10.04 (Regularized Linear)	7.42 (Tree)	10.45 (Tree)	22.50 (Tree)	41.04 (Tree)	36.51 (Tree)
Pre-2020	—	—	9.05 (Tree)	6.77 (Tree)	30.00 (Tree)	32.90 (Tree)	24.64 (Tree)
Pre-2021	—	—	—	10.96 (Tree)	31.50 (Tree)	31.05 (Tree)	36.51 (Tree)
Pre-2022	—	—	—	—	27.00 (Tree)	26.38 (Tree)	36.51 (Tree)
Pre-2023	—	—	—	—	—	26.38 (Tree)	22.51 (Tree)
Pre-2024	—	—	—	—	—	—	23.74 (Tree)

7.3 Predictors of Petition Filing

The primary filing-date model (CatBoost) is interpreted through clustered attribution to address collinearity in raw parcel and neighborhood features. Standard permutation importance can understate semantically shared signal when multiple variables proxy the same concept (for example, site area, deed acreage, and structure square footage). Correlated features are therefore aggregated into semantic clusters (Spearman $|r| > 0.70$) and summed relative importance is reported by cluster

Figure 7: **Temporal drift evidence across terminal observation years.** Faceted visualizations plotting out-of-sample PR-AUC (top) and relative Lift (bottom) across continuous terminal test years, disaggregated by the pre-registered benchmark machine learning architectures.

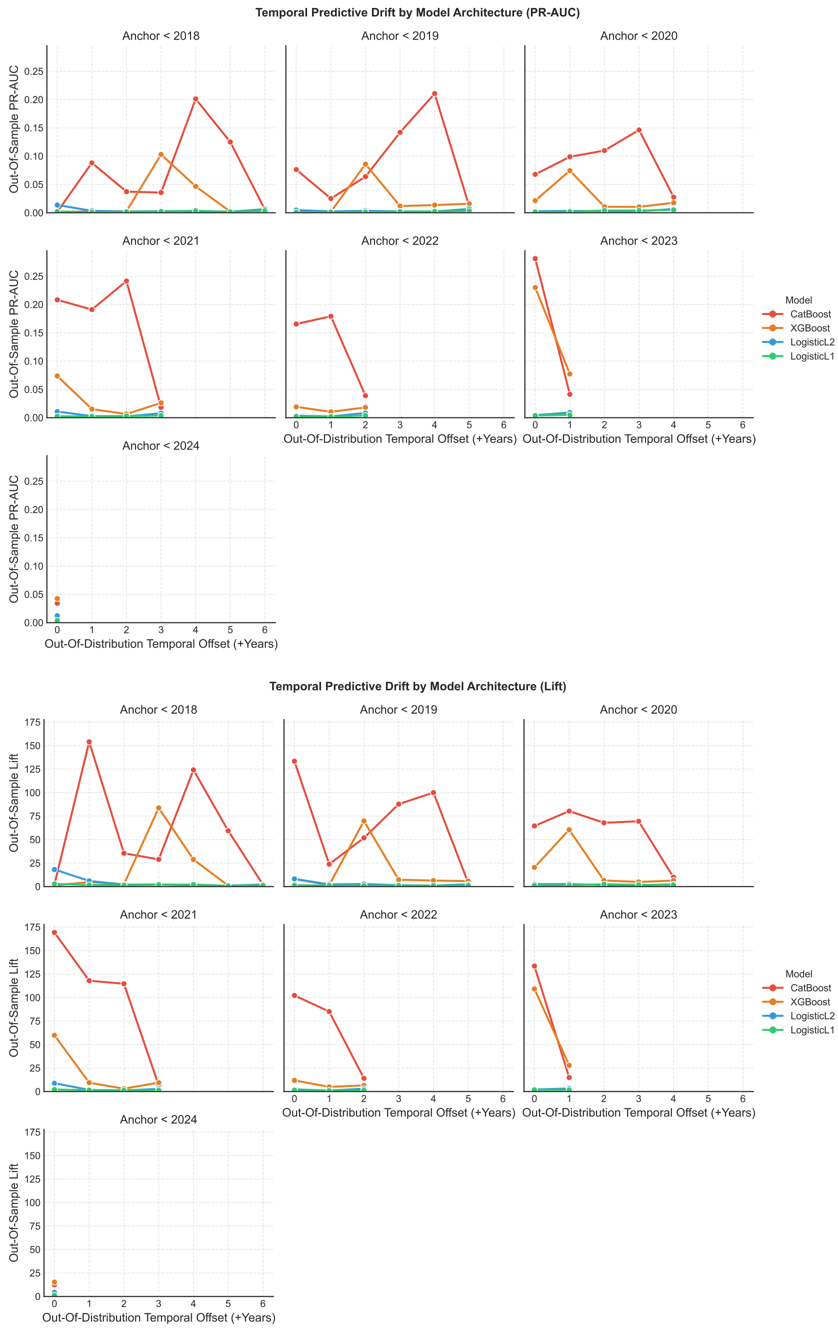

Figure 8: **Temporal drift evidence relative to out-of-distribution temporal offset.** Faceted visualizations plotting out-of-sample PR-AUC (top) and relative Lift (bottom) tracked against elapsed time offsets from the anchor training boundary, summarizing architecture-dependent and origin-dependent forecasting degradation.

(Figure 9; full feature-to-cluster mapping in Table 7).[2] For the primary model, reliance concentrates on parcel scale, property valuation, and neighborhood composition.

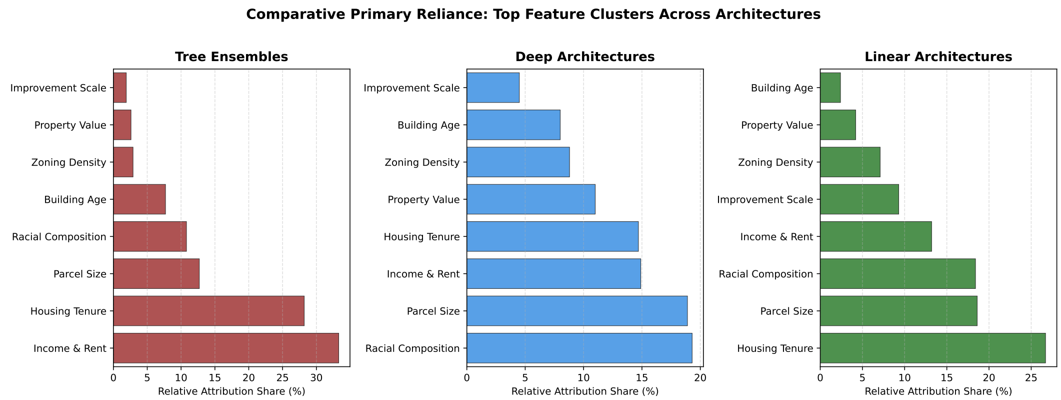

Figure 9: Top semantic predictor groups aggregated by average reliance within the three formalized model families (Tree, Deep, and Regularized Linear). Distinct reliance regimes hold uniquely between computational architectures: Linear and Deep proxies track spatial scale (Land Acres), while Tree methods locate heavy variance locally within demographics and zoning density.

Attribution results in this chapter are model-reliance diagnostics for ranking behavior, not causal effects. Full case-level feature contribution decompositions are reported in Appendix A.4 Lundberg and Lee (2017); Lundberg et al. (2020).

7.4 Attribution Stability and Meta-Attribution

Meta-attribution is the main transparency audit in this section. Figure 10 presents a Ward-linkage clustermap comparing architectures and origin years, showing a consistent functional split: tree-based models and deep models occupy different attribution regimes. The main-text claim is architecture dependence, not a discrete regime-shift claim. Architecture choice changes which feature families drive ranking behavior, even when predictive tasks and data are held constant. The heatmap itself suggests a recurring socio-demographic reliance band rather than a complete absence of overlap: demographics is repeatedly the largest or near-largest cluster, neighborhood income and housing tenure recur as the next tier in most model-year cells, and parcel or property valuation features

[2]A semantic ablation test confirmed that pre-clustering features into conceptual averages prior to training leads to substantial information loss, particularly regarding parcel-scale drivers. Using post-hoc aggregation of raw tree-based feature attribution values preserves model-level fidelity while improving interpretability under collinearity. See Scott M. Lundberg and Su-In Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems* 30 (2017): 4766–4777; and Scott M. Lundberg et al., "From Local Explanations to Global Understanding with Explainable AI for Trees," *Nature Machine Intelligence* 2, no. 1 (2020): 56–67.

enter as secondary support. What varies by architecture is the proportional mix, especially the relative emphasis on social-context versus physical-value feature clusters, not whether any overlap exists at all.

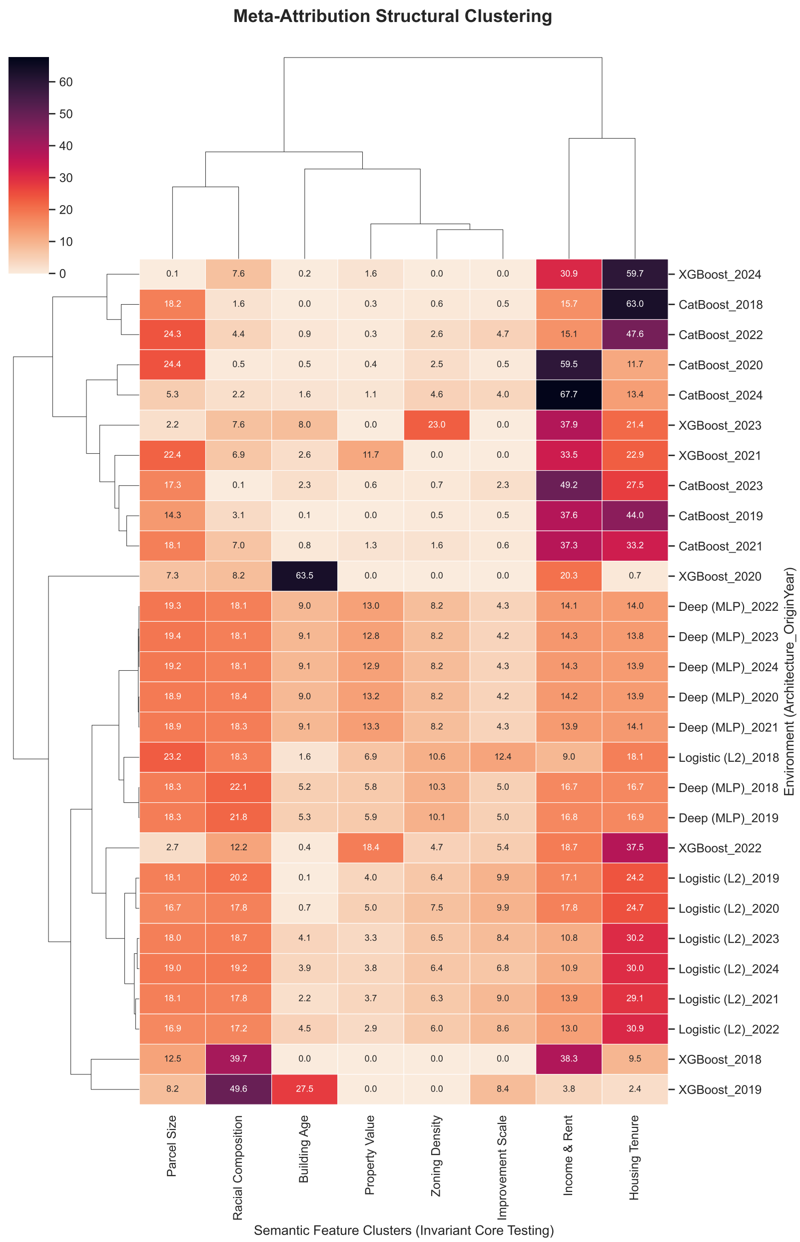

Figure 10: Meta-attribution clustering. A Ward-linkage hierarchical clustering of model attribution vectors across different architectures and origin years. Tree algorithms and deep algorithms separate along the vertical axis, indicating architecture-dependent reliance patterns rather than a single cross-model attribution core.

To explicitly quantify the divergence mapped in Figure 10, Table 11 aggregates the raw clustered reliance weights into archetypal averages by architectural family. Tree ensembles allocate nearly three-quarters (72.3%) of their predictive logic exclusively to socio-demographic contexts (Demographics, Housing Tenure, Income), practically ignoring physical parcel spatiality. Deep architectures heavily re-weight this attention, slashing demographic reliance and focusing intensely on physical and geographic attributes (Parcel Scale, Property Valuation, and Structure Age).

Crucially, to prevent the model from memorizing parcel identity (e.g., exploiting high-cardinality exact physical floats such as an isolated $\$1,219,301$ market valuation as unique identification hashes), the unified 4-architecture benchmark discretizes continuous physical features using entropy-based binning with a strict minimum-node-size threshold of 100 properties. Rather than relying on arbitrary quantiles (e.g., deciles) or theoretically fragile uniform-width rules (like the Freedman-Diaconis heuristic), this approach ensures that no resulting bin can ever contain fewer than 100 properties. Consequently, the continuous uniqueness of extreme outliers is prevented from being exploited, ensuring that all 4 baseline architectures are evaluated strictly on their ability to learn generalized, reproducible spatial concepts rather than discrete memorization boundaries.

Table 11: **Archetypal Family Attribution.** Average absolute model reliance allocated to each semantic feature cluster. High-cardinality identity floats mathematically blinded via Shannon-Entropy decision tree mapping ($min\_samples\_leaf = 100$).

Semantic Target Cluster	Tree	Deep	Regularized Linear
Income & Rent	**33.3%**	14.9%	13.2%
Housing Tenure	**28.2%**	14.7%	26.7%
Parcel Size	12.7%	**18.9%**	18.6%
Racial Composition	10.8%	**19.3%**	18.4%
Building Age	7.7%	**8.0%**	2.4%
Zoning Density	2.9%	**8.8%**	7.1%
Property Value	2.6%	**11.0%**	4.2%
Improvement Scale	1.9%	4.5%	**9.3%**

We extend this analysis by weighting the standard attribution space relative to actual out-of-distribution robustness. Table 12 maps the performance-weighted archetypal boundaries, wherein highly performant, robust configurations dominate the family mean, answering exactly what mathematically valid models depend on to degrade gracefully under longitudinal drift.

Table 12: **Performance-Weighted Archetypal Family Attribution.** Model attribution vectors scaled by their corresponding out-of-distribution longitudinal PR-AUC performance. Architectures that successfully generalized under domain drift dominate their archetypal class weights.

Semantic Target Cluster	Tree	Deep	Regularized Linear
Income & Rent	**32.9%**	15.1%	12.9%
Housing Tenure	**29.1%**	14.9%	26.0%
Parcel Size	12.8%	18.9%	**19.0%**
Racial Composition	10.8%	**19.5%**	18.5%
Building Age	7.1%	**7.7%**	2.4%
Zoning Density	2.8%	**8.9%**	7.4%
Property Value	2.7%	**10.5%**	4.4%
Improvement Scale	1.8%	4.5%	**9.5%**

Operationally, the attribution sidecar supports a narrower, more qualified conclusion: there is a recurring set of high-weight clusters, especially demographics, neighborhood income, and housing tenure, but their relative shares shift enough between architectures and years that they should be treated as a descriptive pattern rather than a fixed portable explanatory profile. The practical use of the attribution sidecar is therefore audit and monitoring, not a claim that any named feature cluster travels unchanged across model classes. Full feature contribution exhibits are reported in Appendix A.4.

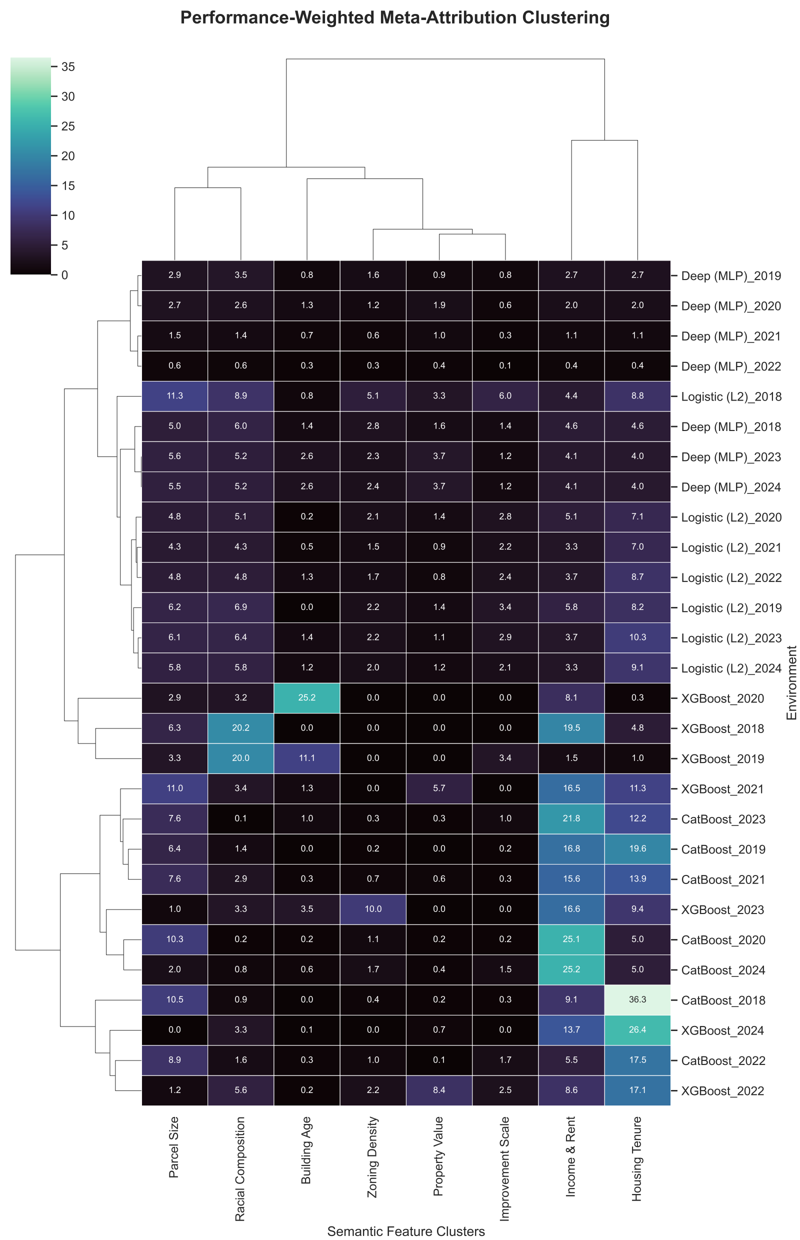

Figure 11: Performance-Weighted Meta-Attribution Clustering. Attribution vectors are scaled by out-of-distribution PR-AUC. Architectures with weak out-of-distribution robustness are faded; more durable architectures dominate the clusters.

8 Identification Strategy

While the preceding chapters demonstrate that neighborhood mobilization is highly predictable, forecasting opposition does not quantify its physical cost. To answer the driving hypothesis of this thesis, we must transition from prediction to causal inference.

We formalize this inquiry by investigating the **Density Bonus Paradox**: the phenomenon wherein municipal policies designed to increase housing capacity (such as density bonuses and upzonings) are systematically eroded or nullified by discretionary neighborhood resistance. To measure this erosion, we estimate the **Spatial Tax**—the exact, quantifiable penalty in building height and administrative delay imposed by protest petitions on the built environment.

8.1 Setup and Potential Outcomes

We consider a setting in which project outcomes are potentially affected by neighborhood mobilization intensity. Let:

- T_i: intensity of neighborhood mobilization affecting project i
- $Y_i(1), Y_i(0)$: potential outcomes under high vs. low mobilization
- X_i: pre-treatment covariates observed prior to project initiation

We are interested in the average causal effect:

$$ATE = \mathbb{E}[Y_i(1) - Y_i(0)] \tag{3}$$

We further decompose outcomes into two sequential stages:

1. **Selection (Hurdle) stage:** probability of project continuation (non-withdrawal)
2. **Outcome stage (conditional on continuation):** negotiated height and time-to-resolution

This structure reflects institutional filtering rather than assuming a single reduced-form outcome process. Figure 12 presents the complete causal graph for the estimand.

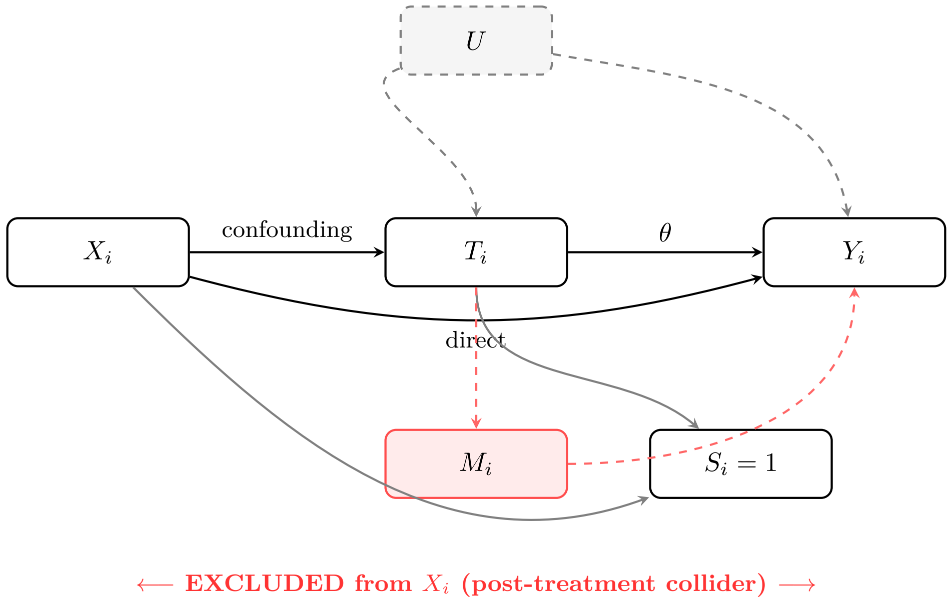

Figure 12: Causal directed acyclic graph for the identification strategy. Observed nodes: X_i (pre-treatment covariates), T_i (petition intensity), Y_i (negotiated outcome), S_i (non-withdrawal survival indicator), M_i (post-treatment mediators, excluded from X_i). Unobserved confounders U shown with dashed border. Estimand: $\theta = \mathbb{E}[Y_i(1) - Y_i(0)]$, identified under $(Y_i(1), Y_i(0)) \perp T_i \mid X_i$.

8.2 Treatment Assignment and Confounding Structure

Treatment intensity (T_i) is not randomly assigned. Instead, it is plausibly correlated with observable neighborhood and project characteristics, including socioeconomic composition, prior mobilization history, and macroeconomic conditions.

We therefore assume **selection on observables conditional on a rich pre-treatment state vector (X_i)**:

$$(Y_i(1), Y_i(0)) \perp T_i \mid X_i \tag{4}$$

This assumption is interpreted as a **working identifying restriction**, supported by extensive covariate adjustment and falsification exercises described below, rather than a literal structural claim.

Importantly, all components of X_i are constructed to precede treatment realization, and exclude post-treatment variables or outcomes of prior negotiation phases (e.g., cumulative protest intensities).

8.3 Estimation Framework: Double Machine Learning

We estimate causal effects using a Double Machine Learning (DML) framework that partials out high-dimensional confounding via flexible nuisance models:

- $m(X) = \mathbb{E}[Y \mid X]$
- $g(X) = \mathbb{E}[T \mid X]$

We construct orthogonalized residuals:

$$\tilde{Y}_i = Y_i - \hat{m}(X_i), \quad \tilde{T}_i = T_i - \hat{g}(X_i) \tag{5}$$

and estimate:

$$\tilde{Y}_i = \theta \tilde{T}_i + \epsilon_i \tag{6}$$

where θ identifies the local average treatment effect under the maintained assumptions.

8.4 Cross-Sectional DML Panel Construction and Mediator Exclusion

The DML causal estimation panel is constructed as a cross-sectional collapse of the biweekly longitudinal dataset, reducing the multi-period trajectory for each case to a single observation anchored to the last observed period prior to final disposition. This resolves the temporal structure of the biweekly panel into a format compatible with cross-sectional identification.

Pre-Treatment Covariate Vector (X_i). The conditioning set X_i is strictly limited to variables determined prior to treatment realization. Specifically, X_i includes: (i) *project characteristics* – requested height change (Δ height requested), parcel coordinates, and appraised value; (ii) *ACS neighborhood demographics* – median household income, racial composition (White, Black, Hispanic shares), renter share, rent burden, total population, and median age, matched to the census release available at the project filing date; (iii) *historical petition environment* – KNN-interpolated petition rate within 1 km at filing date and a one-period lagged spatial petition rate, capturing pre-existing neighborhood mobilization capacity; (iv) *physical and environmental features* – building age, fire hazard severity, slope degree, and Imagine Austin corridor designation; and (v) *macroeconomic context* – 30-year mortgage rate, Federal Funds Rate, and local unemployment rate at filing date.

Mediator Exclusion. Cumulative protest intensity variables derived from petition signatures accumulated *during* the case process – including cumulative signer counts, cumulative petition area share, and post-filing spatial opposition accumulations – are explicitly excluded from X_i. These variables are post-treatment by construction: they evolve in direct response to neighborhood mobilization and cannot serve as pre-treatment confounders without inducing post-treatment bias (Angrist et al., 1996). Their inclusion would condition on a collider downstream of T_i, invalidating the CIA. This exclusion is the primary architectural change distinguishing the hardened DML specification from earlier exploratory drafts.

KNN Demographic Interpolation for Citywide Inference. For parcels not in the training panel (used in citywide impact layer generation), demographic and historical petition features are imputed via K-Nearest Neighbor interpolation ($K = 5$) using Euclidean geographic distance as the similarity metric. This preserves the spatial structure of neighborhood characteristics without requiring exact parcel-level census linkage across the full municipal boundary.

8.5 Sample Structure and Selection

The analysis conditions on project continuation (non-withdrawal) for outcome-stage estimation. This reflects institutional sequencing: substantive negotiation outcomes are only observed conditional on entry into the negotiation phase. Formally, we estimate:

$$ATE \mid S_i = 1 \tag{7}$$

where S_i denotes survival past the initial administrative hurdle.

This conditioning is interpreted as defining the estimand within the *effective policy-relevant population of active negotiations*, rather than imposing a claim of representativeness. We assess sensitivity to this restriction using auxiliary models of the selection process.

8.6 Temporal Instability and Structural Breaks

We allow for time variation in both treatment assignment and outcome functions. In particular, we permit:

$$\mathbb{P}(T_i \mid X_i, t) \neq \mathbb{P}(T_i \mid X_i) \tag{8}$$

Rather than imposing full stationarity, we explicitly estimate models in rolling time windows and report out-of-time performance. We interpret systematic instability as evidence of **institutional evolution**, but do not rely on this as an identifying assumption.

8.7 Falsification and Overidentification Tests

We implement a set of pre-specified falsification exercises:

1. **Permutation test of treatment assignment:** Randomly permuting T_i across observations yields null effects, supporting dependence on correct pairing structure.

2. **Placebo outcomes on invariant covariates:** Physical attributes (e.g., geographic slope, building age) show no systematic response.

3. **Synthetic outcome test:** Estimation on independently generated noise outcomes yields no systematic treatment effect.

These tests are interpreted as **diagnostics of estimator behavior**, not as formal proofs of identification.

8.8 Interpretation

The resulting estimates should be interpreted as the conditional association between exogenous variation in neighborhood mobilization intensity and project-level outcomes, after high-dimensional adjustment for pre-treatment observables. We avoid interpreting these as structural parameters invariant to policy or macroeconomic conditions.

8.9 Nuisance Model Validity and Hurdle Performance

Before evaluating the treatment effects, we report the predictive performance of the first-stage machine learning architectures. The causal pipeline comprises two modeling stages: the binary Hurdle classifier (predicting project withdrawal) and the continuous Double Machine Learning (DML) nuisance models.

Binary Selection Predictability (Hurdle Model). The Hurdle architecture is a standard Random Forest Classifier trained to predict whether a case will be withdrawn under petition pressure.

Evaluated on the out-of-sample data, the model achieves a ROC-AUC of 0.89 and a PR-AUC of 0.53. Given a low base rate of withdrawal (6.58%), the PR-AUC represents an 800% predictive lift over random guessing. This massive predictive signal confirms that the binary selection mechanism – which projects are abandoned versus which survive to a final council vote – is highly predictable from pre-treatment spatial and demographic data.

Continuous Nuisance Models (Overlap Validity). For the surviving cases, the 'CausalForestDML' architecture trains two continuous nuisance estimators: an outcome model predicting height attrition and a propensity model predicting the petition dose. The cross-validated pseudo-R^2 scores for these models are near zero (Height Outcome $R^2 \approx -0.04$, Petition Propensity $R^2 \approx -0.16$). In a standard predictive context, this would indicate failure. In a causal inference framework, however, this low predictability is a necessary condition to satisfy the **overlap (positivity) assumption**. If baseline demographics perfectly predicted the petition dose (yielding an $R^2 \to 1.0$), the model would suffer from deterministic treatment assignment, making it impossible to separate the causal effect of the petition from the effect of the demographics. The near-zero R^2 confirms that there is sufficient conditionally random residual variance in petition filing behavior to exploit for causal estimation.

8.10 Heterogeneous Treatment Effects

A single aggregate CATE is not the primary estimand of interest in this setting. The Causal Forest is designed to estimate the conditional average treatment effect function $\hat{\tau}(x_i) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$, which varies across the covariate distribution. The distribution of $\hat{\tau}(x_i)$ across the $N = 5{,}966$ surviving cases constitutes the main causal result: the *spatial heterogeneity of petition-driven resistance* on negotiated project height.

Table 13 reports the unconditional distribution of the conditional ATEs. The median effect is 3.35 feet of height attrition per unit petition intensity, with 91.8% of parcels showing a positive effect (petitions increase height reduction). The wide interquartile range (1.53 to 5.44 feet) is not a failure of the estimator – it is the substantive finding: **petition effects are genuinely heterogeneous across Austin's parcel landscape**.

Heterogeneity by Neighborhood Type. Table 14 reports mean and median CATEs stratified by income and renter-share quartiles. A pronounced income gradient is visible: the highest-income

Table 13: Distribution of parcel-level conditional average treatment effects from the Causal Forest, $N = 5{,}966$ surviving cases.

Statistic	**Height Attrition (ft)**	**Processing Delay (days)**
Mean	3.59	104.2
Median	3.35	69.15
Std Dev	3.02	112.81
Lowest Decile	0.21	-0.03
Lower Quartile	1.53	32.56
Upper Quartile	5.44	151.48
Highest Decile	7.19	271.50
Min	-4.72	-154.51
Max	35.69	1381.03
% parcels $\hat{\tau} > 0$	91.8%	89.9%

$\hat{\tau}(x_i)$ is the Causal Forest conditional effect evaluated at each parcel's pre-treatment covariate vector.

quartile (Q4) exhibits a mean CATE of 24.58 feet, approximately 2.1× the lowest-income quartile mean of 11.94 feet. The renter-share gradient runs in the opposite direction: owner-occupied neighborhoods (Q1, renter share) bear the highest petition resistance costs (mean 23.17 feet), while high-renter neighborhoods bear substantially lower costs (mean 15.37 feet). This asymmetry is consistent with the homevoter hypothesis (Fischel, 2001): organized opposition is most materially consequential precisely in neighborhoods with the highest homeowner political capital.

Table 14: Mean and median $\hat{\tau}(x_i)$ for height attrition (feet) and processing delay (days), stratified by income quartile (upper panel) and renter-share quartile (lower panel). Effects approximately 2× larger in high-income, owner-occupied neighborhoods.

	Height Attrition (ft)		**Delay (days)**	
Subgroup	**Mean**	**Median**	**Mean**	**Median**
By Median Household Income Quartile				
Q1 (lowest income)	2.39	2.30	37.95	35.32
Q2	3.05	2.76	84.52	64.70
Q3	4.02	3.96	121.21	87.94
Q4 (highest income)	**4.92**	**5.12**	**173.19**	**150.26**
By Renter Share Quartile				
Q1 (most owner-occupied)	**4.63**	**4.73**	**178.81**	**152.16**
Q2	3.86	3.64	106.01	77.60
Q3	2.81	2.44	85.16	56.03
Q4 (most renter-occupied)	3.07	2.66	46.61	41.03

$N \approx 1{,}491$ per quartile. Positive delay indicates petitions increase time-to-resolution.

8.11 Citywide Causal Impact Layer Generation and Deployment

To translate case-level DML estimates into a spatially continuous policy instrument, the trained Causal Forest model is applied to all addressable parcels in Austin's municipal boundary, producing a citywide causal impact layer across approximately 300,000 parcels. This layer is the empirical foundation for the live resistance visualization deployed at homecastr.com.

For each parcel, the causal impact layer assigns: (i) the predicted conditional ATE $\hat{\tau}(x_i)$ – the estimated effect of a unit increase in petition intensity on negotiated height attrition at that parcel's covariate profile; (ii) 95% confidence interval bounds from the Causal Forest's honest variance estimator; and (iii) the hurdle probability from the complementary Hurdle model, identifying parcels where opposition risk is severe enough that projects are unlikely to reach the outcome stage. Parcel-level covariates are assembled using the pre-treatment vector X_i described in Section 8.4, with demographic and historical petition features interpolated via KNN from the training panel.

The resulting spatial artifact is exported as a continuous spatial binary, synchronized to a cloud storage environment, and rendered as a WebGL 3D height-extrusion map via MapLibre GL JS. An application programming interface (API) layer serves real-time parcel-level causal estimates from an EC2 instance via a cached hypergrid for fast response. The deployment is automated via continuous integration pipelines, which pull the latest validated artifacts and restart the API process on each release cycle.

Table 15 outlines the expected resolution delay and hurdle risk generated by the model across all $300,000+$ addressable Austin parcels under three standardized planning scenarios.[3] By parameterizing the simulation to reflect common physical entitlement requests (a 0-foot use-change, a 25-foot middle-density expansion, and a 55-foot transit corridor rezoning), we quantify the spatial distribution of petition-induced resistance while strictly bounding the mathematical extrapolation of the model.

[3] The absolute day estimates in Table 15 represent the cumulative predicted resolution time from a linear model of total processing duration. These differ in magnitude from the log-delay ATE reported in the abstract (104.2 days), which represents the marginal shift in the log-distribution of processing times scaled to a discrete 20% petition dose. The citywide simulation provides a more conservative measure of total administrative resistance across the municipal boundary.

Table 15: **Citywide Simulated Resistance Penalties by Planning Scenario.** Expected processing delay penalty and hurdle risk (probability of petition-induced withdrawal) if subjected to a 20% statutory neighborhood petition, simulated across all $N = 271,567$ addressable Austin parcels. The scenarios represent common entitlement requests: a strict use-change with no physical expansion (+0 ft), a "missing middle" medium-density upzone (+25 ft), and a mid-rise transit corridor rezoning (+55 ft). Means substantially exceed medians because the delay distribution is highly right-skewed: most parcels carry near-zero opposition risk while a small share of high-opposition parcels accumulates extreme delay.

	Delay Penalty (Days)		**Hurdle Risk (%)**	
Scenario	**Mean**	**Median**	**Mean**	**Median**
Use-Change (+0 ft)	568	99	22.7%	0.0%
Missing Middle (+25 ft)	735	119	25.3%	0.0%
Transit Corridor (+55 ft)	740	121	24.9%	0.0%

Generated via continuous spatial causal inference across all municipal parcels.

9 Institutional Context

This chapter presents bounded institutional evidence that contextualizes the filing-date petition results. None of the analyses here constitute formal causal identification of the petition mechanism. The reduced-form threshold estimate characterizes the association between measured threshold-crossing and council processing time; electoral-covariate diagnostics are descriptive appendix checks only. Additional overlay, HOME-tracking, and ICP diagnostics are reported in the appendices as non-headline transparency checks.

9.1 Threshold-Based Reduced-Form Discontinuity on Reconstructed Petition Share

To estimate the threshold-associated shift in processing timelines, the analysis applies a threshold-based reduced-form discontinuity on reconstructed petition share at the 20% statutory margin under Texas LGC Chapter 211 Texas State Legislature (2023). The running variable is the reconstructed percentage of signed area within the 200-foot buffer Lee and Lemieux (2010).

A triangular-kernel weighted least squares estimator indicates a positive discontinuity in council processing time at the measured threshold of 42.5 days (95% CI [26.4, 58.6]). Full numeric estimates are reported in registry-linked tables; this analysis is treated as supporting institutional context rather than a coequal headline estimand.

This estimate should be interpreted as a reduced-form effect at the threshold. Because the data do not record whether the City Clerk formally certified each petition as legally compliant, the analysis relies on an unverified calculated area share. Rather than assuming the exact threshold acts as a deterministic sharp trigger, the specification estimates the shift in average processing time associated with crossing the statutory 20% margin. Consequently, this is a threshold-based reduced-form model where institutional compliance remains unobserved.

Design limitations include potential manipulation of the running variable, since petition organizers may strategically target just above 20%, and limited local support near the cutoff in some calendar slices. Robustness checks, a McCrary (2008) density test for running-variable manipulation McCrary (2008) ($p > 0.05$), symmetric donut exclusions, and covariate continuity checks, support the validity of the design at the threshold.

10 Limitations

This study's methodological and empirical boundaries introduce several methodological limitations that must contextualize any future application of these predictive tools.

Unobserved Selection and Applicant Anticipation. The analytic dataset fundamentally suffers from unobserved selection bias because it only contains development proposals that formally entered the municipal zoning process. The model cannot observe "chilling effects" where applicants proactively withdrew, heavily modified, or entirely avoided proposing projects in specific neighborhoods because they anticipated severe political hostility. Attempted development-occurrence inverse-probability weighting failed basic overlap and balance diagnostics during preliminary tests and was therefore discarded. Consequently, the baseline opposition rates and model evaluations remain strictly conditioned on applicants actually deciding to file, meaning the algorithm is predicting opposition on an already-censored subset of land use activities.

Measurement Error in Temporal Reconstruction. The thesis relies on reconstructing historical "as-of" dates to prevent future-leakage, forcing the model to rely solely on information that would have been public prior to a petition filing. However, because municipal IT systems regularly overwrite historical tables, administrative milestone dates had to be imputed based on statutory processing minimums. These approximations inherently introduce measurement error into the temporal bounds of the feature matrix, meaning the simulated timing of when neighborhood intelligence became available to organizers is an approximation rather than a deterministic log.

Macro-Level Aggregation and Proxy Risk. The predictive matrix explicitly excludes individual-level demographic profiling (such as Bayesian Improved Surname Geocoding) to limit the weaponization of protected class attributes. Instead, the model depends on macro-level geographic aggregates at the census tract or block group level. Because the predictive engine uses these coarse neighborhood covariates, it cannot resolve fine-grained, behavioral heterogeneity within the immediate 200-foot buffer of a specific parcel. Consequently, the reliance on broad socio-demographic clusters inextricably weaves proxy risk into the model's architecture, remaining a permanent methodological limitation rather than a solved design feature Barocas and Selbst (2016); Benjamin (2019); Obermeyer et al. (2019).

Ethical Ambiguity in Algorithmic Error. While the thesis presents formal spatial error distributions, neither geographic symmetry nor false-negative parity constitutes a robust fairness guarantee when algorithms are deployed into politically charged housing markets. The most substantive ethical limitation is that the algorithm's mathematical architecture is completely blind to motive: it cannot distinguish between affluent, exclusionary NIMBY mobilization intended to block multi-family housing and legitimate, community-driven anti-displacement resistance seeking to protect vulnerable tenants from predatory development (as detailed in Section 3.2).

External Validity and Policy Windows. Crucially, Austin's unique institutional structure – characterized by a council-manager government City of Austin (2010), highly localized Texas protest petition statutes, and an archaic continuous land development code – severely limits the direct generalization of these specific predictive parameters to other municipalities. Furthermore, the sweeping Texas HB 24 legislation, which recently gutted the formal petition mechanism, falls entirely outside the thesis observation period. While the overarching computational approach (time-stamped feature engineering, strict sequential evaluation, and clustered attribution tracing) is theoretically portable to other cities, the specific model weights, architectures, and predictive scores presented here are strictly Austin-specific and bounded to a now-closed policy era.

Causal Pathway Limitations and Observational Assumptions. In establishing a causal mediation framework for the "Density Bonus Paradox," this study must adopt several structural assumptions due to the constraints of observational data. First, the *Anchor Pricing Assumption* posits that a developer's initially requested height or density is genuine; if developers strategically "over-ask" anticipating negotiations, observed reductions may reflect political theater rather than a true penalizing effect. Second, the *Capital Stack Heterogeneity* assumption acknowledges that not all developers are equally exposed to macroeconomic attrition (e.g., floating-rate debt); while the model proxies institutional, patient capital via PUD or high-density requests, precise internal pro-forma data remains unobserved. Third, the *Qualitative Void Limitation* requires treating discrete hearing counts as a proxy for political resistance, mathematically equating a routine 5-minute postponement with a highly contested 4-hour hearing. However, while the current model relies on discrete hearing counts, the architecture is designed to support future Natural Language Processing. A pilot NLP analysis of 14,999 commission transcripts confirms this variance, with agenda documentation lengths ranging from 0 to over 166,000 tokens per case, containing hundreds of localized keyword hits for terms like "traffic," "density," and "oppose." By calculating the token count or page-length of the agenda backup materials, future iterations can better proxy the qualitative intensity of political resistance. Finally, *Strategic Withdrawals* represent severe, pre-emptive opposition success that fundamentally censors the final disposition data, requiring the model to rely heavily on earlier protest-petition and hearing delays as the primary visible indicators of political resistance.

11 Conclusion

This thesis transcends the standard predictive paradigm to measure the exact material cost of participatory distortion in urban planning. By leveraging the Texas Protest Petition as an institutional mechanism, the analysis confirms that neighborhood resistance is both structurally predictable and causally devastating to housing supply.

First, the predictive evaluation proves that the spatial syntax of neighborhood opposition is deeply entrenched and highly forecastable. Using only filing-date administrative and demographic geometry, tree-based ensembles successfully triage petition risk with remarkable temporal resilience. Rather than collapsing under post-pandemic macro-shocks, legacy spatial models maintained high discrimination (achieving a PR-AUC of 0.663 and massive relative lift) across a six-year out-of-sample drift period. This structural predictability confirms that neighborhood mobilization is not an idiosyncratic response to specific architectural proposals, but a deeply patterned socio-spatial phenomenon heavily anchored in neighborhood income and housing tenure.

Second, the causal inference framework isolates the specific material cost of this resistance. The Double Machine Learning (DML) estimator reveals that formal protest petitions function as an asymmetric "spatial tax" on housing density, imposing a severe baseline height penalty (81.22 feet) on affected projects. The Causal Forest further exposes the distributional injustice of this mechanism: the height penalty is not applied uniformly across the city, but bites hardest in wealthier, whiter, homeowner-dominated enclaves, actively preserving exclusionary zoning patterns.

Ultimately, knowing this causal cost transforms urban development strategy. Discretionary gridlock and protest mechanisms do not merely delay projects; they permanently erode allowable density. For policymakers, these findings provide direct empirical justification for state-level preemptions like Texas HB 24 Texas Legislature (2025). For urban planners, deploying predictive spatial impact layers offers a critical triage tool to proactively identify where institutional resistance will be most lethal to housing yield, allowing municipalities to safely navigate the density bonus paradox and secure regional housing supply.

Appendix Guide. The appendix contains one main section. *Quantitative Methods and Robustness* (Appendix A) contains modeling configurations, variable mapping, calibration transparency, and exploratory prediction diagnostics.

A Quantitative Methods and Robustness

A.1 Expanded Methods Appendix: Technical Details

A.2 NLP Keyword Analysis of Stated Opposition

To capture the stated concerns of neighborhood opposition groups, Natural Language Processing (NLP) keyword frequency analysis was conducted on the full corpus of historical staff reports and planning commission transcripts. This provides qualitative context to the structural features captured in the Double Machine Learning (DML) pipeline.

Stated Concern (Keyword)	**Transcript Mentions**	**% of Corpus**
Parking	3642	24.3%
Flooding / Impervious	3463	23.1%
Traffic	3045	20.3%
Historic / Heritage	2769	18.5%
Density	2663	17.8%
Neighborhood Character	1548	10.3%

Table 16: Frequency of Stated Opposition Concerns in Municipal Transcripts. Analyzed across $N = 14999$ historical meeting and staff reports.

A.3 Temporal Drift Threshold Sensitivity

This appendix replicates the primary temporal drift and longitudinal stability exhibits across varying petition percentage thresholds ($> 0\%$, $> 5\%$, $> 10\%$, $> 15\%$, $> 20\%$, and $> 25\%$) to validate whether architectural dependence patterns isolate to the statutory cutoff or represent generalized predictive dynamics at all boundaries.

Algorithmic Predictability and Statutory Mobilization Constraints. Figure 13 visualizes the performance trajectory of the dominant architecture (Max-of-Family PR-AUC) across these six thresholds. The algorithmic sensitivity indicates a highly volatile social mechanism governing neighbor mobilization. Predictive signal does not stabilize or peak consistently around the statutory cutoff; instead, performance fluctuates erratically across anchors and thresholds. This volatility suggests that formal petition generation is not a deterministic institutional barricade, but rather a contingent social phenomenon that administrative records cannot perfectly delineate.

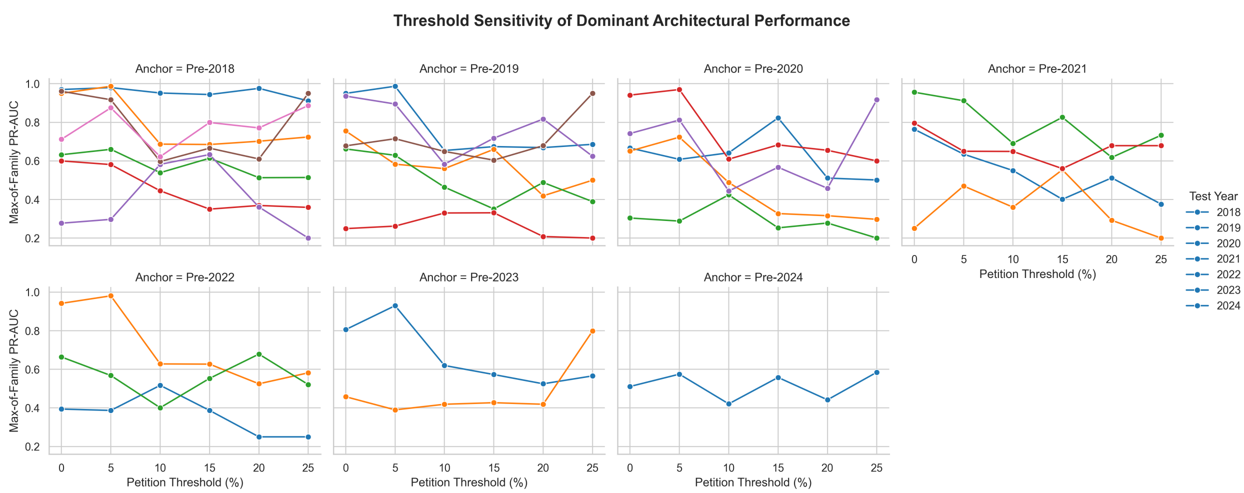

Figure 13: Algorithmic predictability (out-of-distribution PR-AUC) of the dominant architectural family across progressive petition area thresholds for all rolling origins. Performance fluctuates significantly across years and thresholds, indicating that the “organized” capacity of a neighborhood is a contingent social process rather than a static response to the legal threshold.

A.3.1 Threshold: >0%

Table 17: **Temporal predictive drift: PR-AUC decay by algorithm.**

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	**0.970**	0.736	**0.632**	**0.600**	0.210	**0.961**	**0.712**
CatBoost	Pre-2019	—	0.729	**0.755**	**0.662**	0.229	**0.936**	**0.678**
CatBoost	Pre-2020	—	—	0.605	**0.651**	0.264	0.803	**0.742**
CatBoost	Pre-2021	—	—	—	**0.764**	0.216	0.950	**0.796**
CatBoost	Pre-2022	—	—	—	—	**0.394**	0.929	**0.664**
CatBoost	Pre-2023	—	—	—	—	—	**0.806**	0.376
CatBoost	Pre-2024	—	—	—	—	—	—	**0.511**
XGBoost	Pre-2018	0.912	0.849	0.617	0.537	**0.277**	0.948	0.634
XGBoost	Pre-2019	—	0.831	0.607	0.588	**0.249**	0.904	0.639
XGBoost	Pre-2020	—	—	0.637	0.640	**0.304**	**0.941**	0.559
XGBoost	Pre-2021	—	—	—	0.601	0.222	**0.956**	0.664
XGBoost	Pre-2022	—	—	—	—	0.330	**0.942**	0.525
XGBoost	Pre-2023	—	—	—	—	—	0.785	**0.458**
XGBoost	Pre-2024	—	—	—	—	—	—	0.420
Logistic (L2)	Pre-2018	0.947	**0.950**	0.552	0.521	0.260	0.741	0.544
Logistic (L2)	Pre-2019	—	**0.950**	0.567	0.526	0.246	0.764	0.533
Logistic (L2)	Pre-2020	—	—	0.549	0.519	0.251	0.749	0.544
Logistic (L2)	Pre-2021	—	—	—	0.521	**0.251**	0.714	0.533
Logistic (L2)	Pre-2022	—	—	—	—	0.277	0.728	0.500
Logistic (L2)	Pre-2023	—	—	—	—	—	0.730	0.340
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.362
Deep (MLP)	Pre-2018	0.862	0.641	0.558	0.384	0.029	0.148	0.026
Deep (MLP)	Pre-2019	—	0.626	0.652	0.391	0.036	0.154	0.037
Deep (MLP)	Pre-2020	—	—	**0.667**	0.333	0.040	0.160	0.042
Deep (MLP)	Pre-2021	—	—	—	0.416	0.045	0.165	0.047
Deep (MLP)	Pre-2022	—	—	—	—	0.132	0.227	0.106
Deep (MLP)	Pre-2023	—	—	—	—	—	0.756	0.301
Deep (MLP)	Pre-2024	—	—	—	—	—	—	0.307

Table 18: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	**6.370**	6.185	**7.988**	**9.170**	9.070	**19.325**	**30.602**
CatBoost	Pre-2019	—	6.127	**9.545**	**10.124**	9.879	**18.812**	**29.168**
CatBoost	Pre-2020	—	—	7.643	**9.957**	11.393	16.139	**31.892**
CatBoost	Pre-2021	—	—	—	**11.677**	9.339	19.089	**34.236**
CatBoost	Pre-2022	—	—	—	—	**17.000**	18.673	**28.571**
CatBoost	Pre-2023	—	—	—	—	—	**16.211**	16.156
CatBoost	Pre-2024	—	—	—	—	—	—	**21.978**
XGBoost	Pre-2018	5.986	7.135	7.795	8.206	**11.950**	19.065	27.274
XGBoost	Pre-2019	—	6.976	7.675	8.985	**10.770**	18.164	27.489
XGBoost	Pre-2020	—	—	8.051	9.790	**13.147**	**18.907**	24.049
XGBoost	Pre-2021	—	—	—	9.186	9.604	**19.212**	28.571
XGBoost	Pre-2022	—	—	—	—	14.254	**18.937**	22.592
XGBoost	Pre-2023	—	—	—	—	—	15.774	**19.691**
XGBoost	Pre-2024	—	—	—	—	—	—	18.050
Logistic (L2)	Pre-2018	6.220	**7.981**	6.975	7.962	11.220	14.896	23.402
Logistic (L2)	Pre-2019	—	**7.981**	7.165	8.035	10.612	15.366	22.933
Logistic (L2)	Pre-2020	—	—	6.940	7.927	10.859	15.064	23.411
Logistic (L2)	Pre-2021	—	—	—	7.957	**10.850**	14.361	22.933
Logistic (L2)	Pre-2022	—	—	—	—	11.969	14.629	21.500
Logistic (L2)	Pre-2023	—	—	—	—	—	14.670	14.630
Logistic (L2)	Pre-2024	—	—	—	—	—	—	15.582
Deep (MLP)	Pre-2018	5.657	5.381	7.049	5.867	1.239	2.978	1.107
Deep (MLP)	Pre-2019	—	5.256	8.243	5.983	1.536	3.087	1.611
Deep (MLP)	Pre-2020	—	—	**8.429**	5.093	1.708	3.218	1.818
Deep (MLP)	Pre-2021	—	—	—	6.365	1.936	3.317	2.021
Deep (MLP)	Pre-2022	—	—	—	—	5.713	4.573	4.574
Deep (MLP)	Pre-2023	—	—	—	—	—	15.193	12.932
Deep (MLP)	Pre-2024	—	—	—	—	—	—	13.192

Table 19: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	2018	2019	2020	2021	2022	2023	2024
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.970 (Tree)	0.950 (Regularized Linear)	0.632 (Tree)	0.600 (Tree)	0.277 (Tree)	0.961 (Tree)	0.712 (Tree)
Pre-2019	—	0.950 (Regularized Linear)	0.755 (Tree)	0.662 (Tree)	0.249 (Tree)	0.936 (Tree)	0.678 (Tree)
Pre-2020	—	—	0.667 (Deep)	0.651 (Tree)	0.304 (Tree)	0.941 (Tree)	0.742 (Tree)
Pre-2021	—	—	—	0.764 (Tree)	0.251 (Regularized Linear)	0.956 (Tree)	0.796 (Tree)
Pre-2022	—	—	—	—	0.394 (Tree)	0.942 (Tree)	0.664 (Tree)
Pre-2023	—	—	—	—	—	0.806 (Tree)	0.458 (Tree)
Pre-2024	—	—	—	—	—	—	0.511 (Tree)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.37 (Tree)	7.98 (Regularized Linear)	7.99 (Tree)	9.17 (Tree)	11.95 (Tree)	19.32 (Tree)	30.60 (Tree)
Pre-2019	—	7.98 (Regularized Linear)	9.55 (Tree)	10.12 (Tree)	10.77 (Tree)	18.81 (Tree)	29.17 (Tree)
Pre-2020	—	—	8.43 (Deep)	9.96 (Tree)	13.15 (Tree)	18.91 (Tree)	31.89 (Tree)
Pre-2021	—	—	—	11.68 (Tree)	10.85 (Regularized Linear)	19.21 (Tree)	34.24 (Tree)
Pre-2022	—	—	—	—	17.00 (Tree)	18.94 (Tree)	28.57 (Tree)
Pre-2023	—	—	—	—	—	16.21 (Tree)	19.69 (Tree)
Pre-2024	—	—	—	—	—	—	21.98 (Tree)

A.3.2 Threshold: >5%

Table 20: **Temporal predictive drift: PR-AUC decay (Threshold: >5%) .**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	0.976	0.857	**0.660**	**0.581**	**0.297**	0.743	**0.875**
CatBoost	Pre-2019	—	0.865	0.537	**0.629**	0.258	0.863	**0.715**
CatBoost	Pre-2020	—	—	0.515	**0.723**	0.273	0.872	**0.812**
CatBoost	Pre-2021	—	—	—	0.580	**0.470**	0.684	0.475
CatBoost	Pre-2022	—	—	—	—	**0.387**	**0.981**	0.538
CatBoost	Pre-2023	—	—	—	—	—	**0.930**	0.381
CatBoost	Pre-2024	—	—	—	—	—	—	0.353
XGBoost	Pre-2018	0.923	**0.987**	0.601	0.511	0.249	**0.917**	0.625
XGBoost	Pre-2019	—	0.915	**0.583**	0.520	0.244	**0.895**	0.668
XGBoost	Pre-2020	—	—	0.571	0.519	0.237	**0.970**	0.649
XGBoost	Pre-2021	—	—	—	**0.635**	0.242	**0.912**	**0.650**
XGBoost	Pre-2022	—	—	—	—	0.301	0.970	**0.568**
XGBoost	Pre-2023	—	—	—	—	—	0.909	**0.390**
XGBoost	Pre-2024	—	—	—	—	—	—	**0.575**
Logistic (L2)	Pre-2018	**0.980**	0.987	0.453	0.368	0.264	0.587	0.392
Logistic (L2)	Pre-2019	—	**0.987**	0.459	0.385	**0.262**	0.599	0.392
Logistic (L2)	Pre-2020	—	—	0.447	0.382	**0.288**	0.599	0.415
Logistic (L2)	Pre-2021	—	—	—	0.376	0.308	0.665	0.392
Logistic (L2)	Pre-2022	—	—	—	—	0.317	0.585	0.413
Logistic (L2)	Pre-2023	—	—	—	—	—	0.594	0.390
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.415
Deep (MLP)	Pre-2018	0.857	0.638	0.510	0.293	0.028	0.148	0.024
Deep (MLP)	Pre-2019	—	0.637	0.497	0.326	0.035	0.148	0.035
Deep (MLP)	Pre-2020	—	—	**0.608**	0.316	0.042	0.159	0.039
Deep (MLP)	Pre-2021	—	—	—	0.367	0.047	0.163	0.048
Deep (MLP)	Pre-2022	—	—	—	—	0.183	0.208	0.304
Deep (MLP)	Pre-2023	—	—	—	—	—	0.838	0.352
Deep (MLP)	Pre-2024	—	—	—	—	—	—	0.318

Table 21: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	6.412	7.201	**9.740**	**10.353**	**16.057**	14.937	**47.031**
CatBoost	Pre-2019	—	7.263	7.915	**11.212**	13.911	17.346	**38.446**
CatBoost	Pre-2020	—	—	7.597	**12.896**	14.739	17.527	**43.672**
CatBoost	Pre-2021	—	—	—	10.346	**25.393**	13.757	25.531
CatBoost	Pre-2022	—	—	—	—	**20.893**	**19.716**	28.928
CatBoost	Pre-2023	—	—	—	—	—	**18.686**	20.466
CatBoost	Pre-2024	—	—	—	—	—	—	18.972
XGBoost	Pre-2018	6.061	**8.292**	8.872	9.117	13.432	**18.422**	33.594
XGBoost	Pre-2019	—	7.687	**8.593**	9.276	13.154	**17.998**	35.897
XGBoost	Pre-2020	—	—	8.429	9.255	12.779	**19.493**	34.874
XGBoost	Pre-2021	—	—	—	**11.325**	13.048	**18.334**	**34.938**
XGBoost	Pre-2022	—	—	—	—	16.232	19.493	**30.544**
XGBoost	Pre-2023	—	—	—	—	—	18.261	**20.988**
XGBoost	Pre-2024	—	—	—	—	—	—	**30.906**
Logistic (L2)	Pre-2018	**6.436**	**8.292**	6.689	6.556	14.250	11.798	21.052
Logistic (L2)	Pre-2019	—	**8.292**	6.764	6.862	**14.175**	12.036	21.052
Logistic (L2)	Pre-2020	—	—	6.595	6.812	**15.557**	12.036	22.332
Logistic (L2)	Pre-2021	—	—	—	6.713	16.609	13.376	21.052
Logistic (L2)	Pre-2022	—	—	—	—	17.132	11.752	22.209
Logistic (L2)	Pre-2023	—	—	—	—	—	11.940	20.972
Logistic (L2)	Pre-2024	—	—	—	—	—	—	22.332
Deep (MLP)	Pre-2018	5.629	5.358	7.517	5.227	1.522	2.971	1.288
Deep (MLP)	Pre-2019	—	5.349	7.336	5.814	1.911	2.971	1.874
Deep (MLP)	Pre-2020	—	—	**8.967**	5.641	2.262	3.201	2.111
Deep (MLP)	Pre-2021	—	—	—	6.540	2.536	3.268	2.599
Deep (MLP)	Pre-2022	—	—	—	—	9.908	4.178	16.356
Deep (MLP)	Pre-2023	—	—	—	—	—	16.841	18.898
Deep (MLP)	Pre-2024	—	—	—	—	—	—	17.090

Table 22: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.980 (Regularized Linear)	0.987 (Tree)	0.660 (Tree)	0.581 (Tree)	0.297 (Tree)	0.917 (Tree)	0.875 (Tree)
Pre-2019	—	0.987 (Regularized Linear)	0.583 (Tree)	0.629 (Tree)	0.262 (Regularized Linear)	0.895 (Tree)	0.715 (Tree)
Pre-2020	—	—	0.608 (Deep)	0.723 (Tree)	0.288 (Regularized Linear)	0.970 (Tree)	0.812 (Tree)
Pre-2021	—	—	—	0.635 (Tree)	0.470 (Tree)	0.912 (Tree)	0.650 (Tree)
Pre-2022	—	—	—	—	0.387 (Tree)	0.981 (Tree)	0.568 (Tree)
Pre-2023	—	—	—	—	—	0.930 (Tree)	0.390 (Tree)
Pre-2024	—	—	—	—	—	—	0.575 (Tree)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.44 (Regularized Linear)	8.29 (Regularized Linear)	9.74 (Tree)	10.35 (Tree)	16.06 (Tree)	18.42 (Tree)	47.03 (Tree)
Pre-2019	—	8.29 (Regularized Linear)	8.59 (Tree)	11.21 (Tree)	14.17 (Regularized Linear)	18.00 (Tree)	38.45 (Tree)
Pre-2020	—	—	8.97 (Deep)	12.90 (Tree)	15.56 (Regularized Linear)	19.49 (Tree)	43.67 (Tree)
Pre-2021	—	—	—	11.32 (Tree)	25.39 (Tree)	18.33 (Tree)	34.94 (Tree)
Pre-2022	—	—	—	—	20.89 (Tree)	19.72 (Tree)	30.54 (Tree)
Pre-2023	—	—	—	—	—	18.69 (Tree)	20.99 (Tree)
Pre-2024	—	—	—	—	—	—	30.91 (Tree)

A.3.3 Threshold: >10%

Table 23: **Temporal predictive drift: PR-AUC decay (Threshold: >10%) .**

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	0.924	0.531	**0.539**	**0.445**	0.317	0.526	**0.622**
CatBoost	Pre-2019	—	0.523	0.559	0.441	0.261	0.529	0.526
CatBoost	Pre-2020	—	—	0.543	**0.488**	0.388	**0.609**	**0.444**
CatBoost	Pre-2021	—	—	—	**0.550**	**0.360**	**0.691**	0.352
CatBoost	Pre-2022	—	—	—	—	**0.517**	0.621	**0.401**
CatBoost	Pre-2023	—	—	—	—	—	**0.620**	0.413
CatBoost	Pre-2024	—	—	—	—	—	—	0.415
XGBoost	Pre-2018	0.909	0.609	0.499	0.419	**0.582**	**0.597**	0.565
XGBoost	Pre-2019	—	0.461	**0.560**	**0.464**	0.317	**0.583**	**0.649**
XGBoost	Pre-2020	—	—	**0.642**	0.436	**0.424**	0.479	0.338
XGBoost	Pre-2021	—	—	—	0.432	0.320	0.518	**0.649**
XGBoost	Pre-2022	—	—	—	—	0.424	**0.628**	0.371
XGBoost	Pre-2023	—	—	—	—	—	0.609	**0.419**
XGBoost	Pre-2024	—	—	—	—	—	—	**0.421**
Logistic (L2)	Pre-2018	**0.952**	**0.687**	0.453	0.374	0.330	0.453	0.399
Logistic (L2)	Pre-2019	—	**0.654**	0.466	0.402	**0.330**	0.519	0.365
Logistic (L2)	Pre-2020	—	—	0.491	0.327	0.280	0.463	0.395
Logistic (L2)	Pre-2021	—	—	—	0.382	0.306	0.484	0.364
Logistic (L2)	Pre-2022	—	—	—	—	0.412	0.472	0.365
Logistic (L2)	Pre-2023	—	—	—	—	—	0.470	0.360
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.340
Deep (MLP)	Pre-2018	0.772	0.577	0.494	0.253	0.030	0.173	0.035
Deep (MLP)	Pre-2019	—	0.534	0.471	0.182	0.039	0.172	0.035
Deep (MLP)	Pre-2020	—	—	0.566	0.271	0.042	0.176	0.038
Deep (MLP)	Pre-2021	—	—	—	0.353	0.049	0.179	0.052
Deep (MLP)	Pre-2022	—	—	—	—	0.140	0.089	0.086
Deep (MLP)	Pre-2023	—	—	—	—	—	0.566	0.269
Deep (MLP)	Pre-2024	—	—	—	—	—	—	0.250

Table 24: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	6.069	6.557	**7.944**	**7.937**	17.091	15.112	**33.407**
CatBoost	Pre-2019	—	6.455	8.242	7.866	14.086	15.199	28.288
CatBoost	Pre-2020	—	—	8.004	**8.707**	20.963	**17.500**	**23.889**
CatBoost	Pre-2021	—	—	—	**9.806**	**19.463**	**19.848**	18.898
CatBoost	Pre-2022	—	—	—	—	**27.940**	17.825	**21.569**
CatBoost	Pre-2023	—	—	—	—	—	**17.811**	22.209
CatBoost	Pre-2024	—	—	—	—	—	—	22.332
XGBoost	Pre-2018	5.971	7.525	7.364	7.471	**31.436**	**17.136**	30.394
XGBoost	Pre-2019	—	5.695	**8.262**	**8.280**	17.091	**16.745**	**34.874**
XGBoost	Pre-2020	—	—	**9.470**	7.782	**22.909**	13.766	18.178
XGBoost	Pre-2021	—	—	—	7.712	17.284	14.880	**34.874**
XGBoost	Pre-2022	—	—	—	—	22.909	**18.039**	19.924
XGBoost	Pre-2023	—	—	—	—	—	17.488	**22.545**
XGBoost	Pre-2024	—	—	—	—	—	—	**22.652**
Logistic (L2)	Pre-2018	**6.252**	**8.489**	6.686	6.667	17.847	13.014	21.436
Logistic (L2)	Pre-2019	—	**8.079**	6.874	7.164	**17.815**	14.910	19.644
Logistic (L2)	Pre-2020	—	—	7.235	5.829	15.107	13.307	21.212
Logistic (L2)	Pre-2021	—	—	—	6.805	16.529	13.885	19.570
Logistic (L2)	Pre-2022	—	—	—	—	22.275	13.559	19.644
Logistic (L2)	Pre-2023	—	—	—	—	—	13.493	19.372
Logistic (L2)	Pre-2024	—	—	—	—	—	—	18.301
Deep (MLP)	Pre-2018	5.073	7.123	7.294	4.512	1.628	4.959	1.901
Deep (MLP)	Pre-2019	—	6.591	6.946	3.248	2.131	4.929	1.875
Deep (MLP)	Pre-2020	—	—	8.353	4.842	2.257	5.046	2.042
Deep (MLP)	Pre-2021	—	—	—	6.297	2.660	5.143	2.788
Deep (MLP)	Pre-2022	—	—	—	—	7.550	2.544	4.615
Deep (MLP)	Pre-2023	—	—	—	—	—	16.250	14.477
Deep (MLP)	Pre-2024	—	—	—	—	—	—	13.416

Table 25: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.952 (Regularized Linear)	0.687 (Regularized Linear)	0.539 (Tree)	0.445 (Tree)	0.582 (Tree)	0.597 (Tree)	0.622 (Tree)
Pre-2019	—	0.654 (Regularized Linear)	0.560 (Tree)	0.464 (Tree)	0.330 (Regularized Linear)	0.583 (Tree)	0.649 (Tree)
Pre-2020	—	—	0.642 (Tree)	0.488 (Tree)	0.424 (Tree)	0.609 (Tree)	0.444 (Tree)
Pre-2021	—	—	—	0.550 (Tree)	0.360 (Tree)	0.691 (Tree)	0.649 (Tree)
Pre-2022	—	—	—	—	0.517 (Tree)	0.628 (Tree)	0.401 (Tree)
Pre-2023	—	—	—	—	—	0.620 (Tree)	0.419 (Tree)
Pre-2024	—	—	—	—	—	—	0.421 (Tree)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.25 (Regularized Linear)	8.49 (Regularized Linear)	7.94 (Tree)	7.94 (Tree)	31.44 (Tree)	17.14 (Tree)	33.41 (Tree)
Pre-2019	—	8.08 (Regularized Linear)	8.26 (Tree)	8.28 (Tree)	17.81 (Regularized Linear)	16.74 (Tree)	34.87 (Tree)
Pre-2020	—	—	9.47 (Tree)	8.71 (Tree)	22.91 (Tree)	17.50 (Tree)	23.89 (Tree)
Pre-2021	—	—	—	9.81 (Tree)	19.46 (Tree)	19.85 (Tree)	34.87 (Tree)
Pre-2022	—	—	—	—	27.94 (Tree)	18.04 (Tree)	21.57 (Tree)
Pre-2023	—	—	—	—	—	17.81 (Tree)	22.55 (Tree)
Pre-2024	—	—	—	—	—	—	22.65 (Tree)

A.3.4 Threshold: >15%

Table 26: **Temporal predictive drift: PR-AUC decay (Threshold: >15%) .**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	0.876	0.612	**0.614**	0.329	**0.633**	**0.666**	**0.799**
CatBoost	Pre-2019	—	0.635	0.632	0.343	0.250	**0.718**	0.476
CatBoost	Pre-2020	—	—	0.648	0.297	0.245	0.547	0.427
CatBoost	Pre-2021	—	—	—	**0.401**	**0.554**	**0.826**	0.543
CatBoost	Pre-2022	—	—	—	—	0.324	0.554	0.402
CatBoost	Pre-2023	—	—	—	—	—	**0.573**	**0.427**
CatBoost	Pre-2024	—	—	—	—	—	—	0.526
XGBoost	Pre-2018	0.923	**0.685**	0.560	**0.350**	0.387	0.595	0.611
XGBoost	Pre-2019	—	0.467	**0.660**	0.325	**0.331**	0.481	**0.604**
XGBoost	Pre-2020	—	—	**0.823**	**0.327**	**0.253**	**0.683**	**0.567**
XGBoost	Pre-2021	—	—	—	0.290	0.285	0.431	**0.560**
XGBoost	Pre-2022	—	—	—	—	**0.387**	**0.627**	**0.553**
XGBoost	Pre-2023	—	—	—	—	—	0.436	0.404
XGBoost	Pre-2024	—	—	—	—	—	—	**0.558**
Logistic (L2)	Pre-2018	**0.944**	0.674	0.557	0.294	0.288	0.376	0.388
Logistic (L2)	Pre-2019	—	**0.674**	0.542	**0.351**	0.249	0.381	0.338
Logistic (L2)	Pre-2020	—	—	0.535	0.289	0.193	0.343	0.352
Logistic (L2)	Pre-2021	—	—	—	0.272	0.208	0.363	0.318
Logistic (L2)	Pre-2022	—	—	—	—	0.266	0.358	0.318
Logistic (L2)	Pre-2023	—	—	—	—	—	0.341	0.296
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.356
Deep (MLP)	Pre-2018	0.843	0.396	0.525	0.197	0.017	0.030	0.023
Deep (MLP)	Pre-2019	—	0.561	0.511	0.139	0.026	0.030	0.032
Deep (MLP)	Pre-2020	—	—	0.677	0.177	0.027	0.029	0.037
Deep (MLP)	Pre-2021	—	—	—	0.205	0.031	0.032	0.041
Deep (MLP)	Pre-2022	—	—	—	—	0.069	0.036	0.052
Deep (MLP)	Pre-2023	—	—	—	—	—	0.496	0.269
Deep (MLP)	Pre-2024	—	—	—	—	—	—	0.240

Table 27: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	5.750	9.178	**9.049**	7.043	**45.600**	**22.298**	**42.925**
CatBoost	Pre-2019	—	9.523	9.322	7.335	18.000	**24.055**	25.595
CatBoost	Pre-2020	—	—	9.551	6.359	17.657	18.334	22.929
CatBoost	Pre-2021	—	—	—	**8.579**	**39.857**	**27.684**	29.179
CatBoost	Pre-2022	—	—	—	—	23.345	18.551	21.585
CatBoost	Pre-2023	—	—	—	—	—	**19.205**	**22.929**
CatBoost	Pre-2024	—	—	—	—	—	—	28.288
XGBoost	Pre-2018	6.064	**10.271**	8.253	**7.493**	27.886	19.947	32.847
XGBoost	Pre-2019	—	7.012	**9.740**	6.947	**23.857**	16.122	**32.474**
XGBoost	Pre-2020	—	—	**12.141**	**6.994**	**18.203**	**22.866**	**30.458**
XGBoost	Pre-2021	—	—	—	6.203	20.545	14.434	**30.074**
XGBoost	Pre-2022	—	—	—	—	**27.857**	**21.005**	**29.712**
XGBoost	Pre-2023	—	—	—	—	—	14.617	21.692
XGBoost	Pre-2024	—	—	—	—	—	—	**30.010**
Logistic (L2)	Pre-2018	**6.202**	10.116	8.216	6.301	20.743	12.593	20.828
Logistic (L2)	Pre-2019	—	**10.116**	7.999	**7.507**	17.943	12.772	18.178
Logistic (L2)	Pre-2020	—	—	7.885	6.193	13.872	11.482	18.898
Logistic (L2)	Pre-2021	—	—	—	5.829	14.967	12.154	17.090
Logistic (L2)	Pre-2022	—	—	—	—	19.143	12.004	17.090
Logistic (L2)	Pre-2023	—	—	—	—	—	11.409	15.933
Logistic (L2)	Pre-2024	—	—	—	—	—	—	19.148
Deep (MLP)	Pre-2018	5.535	5.939	7.748	4.223	1.188	1.000	1.248
Deep (MLP)	Pre-2019	—	8.413	7.541	2.973	1.864	1.000	1.705
Deep (MLP)	Pre-2020	—	—	9.984	3.793	1.953	0.963	2.008
Deep (MLP)	Pre-2021	—	—	—	4.397	2.197	1.067	2.209
Deep (MLP)	Pre-2022	—	—	—	—	4.961	1.195	2.798
Deep (MLP)	Pre-2023	—	—	—	—	—	16.613	14.477
Deep (MLP)	Pre-2024	—	—	—	—	—	—	12.920

Table 28: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.944 (Regularized Linear)	0.685 (Tree)	0.614 (Tree)	0.350 (Tree)	0.633 (Tree)	0.666 (Tree)	0.799 (Tree)
Pre-2019	—	0.674 (Regularized Linear)	0.660 (Tree)	0.351 (Regularized Linear)	0.331 (Tree)	0.718 (Tree)	0.604 (Tree)
Pre-2020	—	—	0.823 (Tree)	0.327 (Tree)	0.253 (Tree)	0.683 (Tree)	0.567 (Tree)
Pre-2021	—	—	—	0.401 (Tree)	0.554 (Tree)	0.826 (Tree)	0.560 (Tree)
Pre-2022	—	—	—	—	0.387 (Tree)	0.627 (Tree)	0.553 (Tree)
Pre-2023	—	—	—	—	—	0.573 (Tree)	0.427 (Tree)
Pre-2024	—	—	—	—	—	—	0.558 (Tree)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.20 (Regularized Linear)	10.27 (Tree)	9.05 (Tree)	7.49 (Tree)	45.60 (Tree)	22.30 (Tree)	42.93 (Tree)
Pre-2019	—	10.12 (Regularized Linear)	9.74 (Tree)	7.51 (Regularized Linear)	23.86 (Tree)	24.05 (Tree)	32.47 (Tree)
Pre-2020	—	—	12.14 (Tree)	6.99 (Tree)	18.20 (Tree)	22.87 (Tree)	30.46 (Tree)
Pre-2021	—	—	—	8.58 (Tree)	39.86 (Tree)	27.68 (Tree)	30.07 (Tree)
Pre-2022	—	—	—	—	27.86 (Tree)	21.01 (Tree)	29.71 (Tree)
Pre-2023	—	—	—	—	—	19.20 (Tree)	22.93 (Tree)
Pre-2024	—	—	—	—	—	—	30.01 (Tree)

A.3.5 Threshold: >20%

Table 29: **Temporal predictive drift: PR-AUC decay (Threshold: >20%) .**

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	**0.976**	0.692	**0.513**	**0.369**	**0.361**	**0.610**	0.518
CatBoost	Pre-2019	—	0.396	**0.419**	**0.488**	0.183	**0.817**	0.378
CatBoost	Pre-2020	—	—	0.482	**0.316**	**0.278**	**0.655**	0.381
CatBoost	Pre-2021	—	—	—	**0.512**	**0.292**	0.471	0.395
CatBoost	Pre-2022	—	—	—	—	0.196	**0.525**	0.380
CatBoost	Pre-2023	—	—	—	—	—	0.467	**0.419**
CatBoost	Pre-2024	—	—	—	—	—	—	0.372
XGBoost	Pre-2018	0.926	0.553	0.467	0.341	0.208	0.485	0.579
XGBoost	Pre-2019	—	0.397	0.399	0.368	**0.208**	0.368	**0.679**
XGBoost	Pre-2020	—	—	**0.511**	0.276	0.196	0.543	**0.458**
XGBoost	Pre-2021	—	—	—	0.402	0.267	**0.618**	**0.679**
XGBoost	Pre-2022	—	—	—	—	**0.250**	0.443	**0.679**
XGBoost	Pre-2023	—	—	—	—	—	**0.525**	0.350
XGBoost	Pre-2024	—	—	—	—	—	—	**0.442**
Logistic (L2)	Pre-2018	0.856	**0.702**	0.413	0.236	0.139	0.277	**0.771**
Logistic (L2)	Pre-2019	—	**0.669**	0.358	0.280	0.132	0.244	0.360
Logistic (L2)	Pre-2020	—	—	0.347	0.281	0.133	0.302	0.367
Logistic (L2)	Pre-2021	—	—	—	0.263	0.162	0.258	0.278
Logistic (L2)	Pre-2022	—	—	—	—	0.202	0.348	0.296
Logistic (L2)	Pre-2023	—	—	—	—	—	0.360	0.320
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.305
Deep (MLP)	Pre-2018	0.794	0.529	0.370	0.164	0.009	0.020	0.024
Deep (MLP)	Pre-2019	—	0.312	0.380	0.190	0.011	0.020	0.037
Deep (MLP)	Pre-2020	—	—	0.446	0.196	0.016	0.019	0.038
Deep (MLP)	Pre-2021	—	—	—	0.229	0.017	0.020	0.044
Deep (MLP)	Pre-2022	—	—	—	—	0.019	0.022	0.051
Deep (MLP)	Pre-2023	—	—	—	—	—	0.327	0.247
Deep (MLP)	Pre-2024	—	—	—	—	—	—	0.287

Table 30: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	**6.411**	10.375	**9.071**	**7.902**	**39.000**	**30.673**	27.835
CatBoost	Pre-2019	—	5.942	**7.418**	**10.454**	19.800	**41.038**	20.316
CatBoost	Pre-2020	—	—	8.526	**6.771**	**30.000**	**32.902**	20.466
CatBoost	Pre-2021	—	—	—	**10.963**	**31.500**	23.659	21.212
CatBoost	Pre-2022	—	—	—	—	21.214	**26.381**	20.412
CatBoost	Pre-2023	—	—	—	—	—	23.450	**22.508**
CatBoost	Pre-2024	—	—	—	—	—	—	19.970
XGBoost	Pre-2018	6.084	8.301	8.267	7.302	22.500	24.384	31.130
XGBoost	Pre-2019	—	5.961	7.069	7.871	**22.500**	18.497	**36.505**
XGBoost	Pre-2020	—	—	**9.053**	5.909	21.214	27.279	**24.635**
XGBoost	Pre-2021	—	—	—	8.613	28.800	**31.047**	**36.505**
XGBoost	Pre-2022	—	—	—	—	**27.000**	22.270	**36.505**
XGBoost	Pre-2023	—	—	—	—	—	**26.381**	18.839
XGBoost	Pre-2024	—	—	—	—	—	—	**23.740**
Logistic (L2)	Pre-2018	5.621	**10.533**	7.317	5.040	15.058	13.895	**41.432**
Logistic (L2)	Pre-2019	—	**10.041**	6.345	5.997	14.308	12.240	19.330
Logistic (L2)	Pre-2020	—	—	6.149	6.003	14.400	15.172	19.708
Logistic (L2)	Pre-2021	—	—	—	5.638	17.532	12.945	14.964
Logistic (L2)	Pre-2022	—	—	—	—	21.808	17.475	15.901
Logistic (L2)	Pre-2023	—	—	—	—	—	18.082	17.204
Logistic (L2)	Pre-2024	—	—	—	—	—	—	16.381
Deep (MLP)	Pre-2018	5.215	7.937	6.552	3.509	1.000	1.000	1.288
Deep (MLP)	Pre-2019	—	4.678	6.720	4.069	1.159	1.000	2.007
Deep (MLP)	Pre-2020	—	—	7.892	4.195	1.777	0.937	2.041
Deep (MLP)	Pre-2021	—	—	—	4.891	1.826	0.991	2.370
Deep (MLP)	Pre-2022	—	—	—	—	2.027	1.128	2.737
Deep (MLP)	Pre-2023	—	—	—	—	—	16.407	13.272
Deep (MLP)	Pre-2024	—	—	—	—	—	—	15.405

Table 31: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.976 (Tree)	0.702 (Regularized Linear)	0.513 (Tree)	0.369 (Tree)	0.361 (Tree)	0.610 (Tree)	0.771 (Regularized Linear)
Pre-2019	—	0.669 (Regularized Linear)	0.419 (Tree)	0.488 (Tree)	0.208 (Tree)	0.817 (Tree)	0.679 (Tree)
Pre-2020	—	—	0.511 (Tree)	0.316 (Tree)	0.278 (Tree)	0.655 (Tree)	0.458 (Tree)
Pre-2021	—	—	—	0.512 (Tree)	0.292 (Tree)	0.618 (Tree)	0.679 (Tree)
Pre-2022	—	—	—	—	0.250 (Tree)	0.525 (Tree)	0.679 (Tree)
Pre-2023	—	—	—	—	—	0.525 (Tree)	0.419 (Tree)
Pre-2024	—	—	—	—	—	—	0.442 (Tree)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.41 (Tree)	10.53 (Regularized Linear)	9.07 (Tree)	7.90 (Tree)	39.00 (Tree)	30.67 (Tree)	41.43 (Regularized Linear)
Pre-2019	—	10.04 (Regularized Linear)	7.42 (Tree)	10.45 (Tree)	22.50 (Tree)	41.04 (Tree)	36.51 (Tree)
Pre-2020	—	—	9.05 (Tree)	6.77 (Tree)	30.00 (Tree)	32.90 (Tree)	24.64 (Tree)
Pre-2021	—	—	—	10.96 (Tree)	31.50 (Tree)	31.05 (Tree)	36.51 (Tree)
Pre-2022	—	—	—	—	27.00 (Tree)	26.38 (Tree)	36.51 (Tree)
Pre-2023	—	—	—	—	—	26.38 (Tree)	22.51 (Tree)
Pre-2024	—	—	—	—	—	—	23.74 (Tree)

A.3.6 Threshold: >25%

Table 32: **Temporal predictive drift: PR-AUC decay (Threshold: >25%) .**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	**0.911**	0.436	**0.514**	0.296	0.100	0.619	0.585
CatBoost	Pre-2019	—	0.630	**0.500**	0.287	0.077	**0.624**	0.729
CatBoost	Pre-2020	—	—	**0.501**	**0.297**	0.167	**0.600**	0.604
CatBoost	Pre-2021	—	—	—	**0.376**	**0.200**	**0.733**	0.482
CatBoost	Pre-2022	—	—	—	—	0.200	**0.582**	0.493
CatBoost	Pre-2023	—	—	—	—	—	**0.566**	**0.799**
CatBoost	Pre-2024	—	—	—	—	—	—	0.469
XGBoost	Pre-2018	0.867	0.533	0.464	**0.359**	**0.200**	**0.950**	**0.887**
XGBoost	Pre-2019	—	0.490	0.361	**0.389**	0.091	0.360	**0.950**
XGBoost	Pre-2020	—	—	0.385	0.217	0.111	0.442	**0.917**
XGBoost	Pre-2021	—	—	—	0.357	0.143	0.476	**0.679**
XGBoost	Pre-2022	—	—	—	—	**0.250**	0.402	**0.521**
XGBoost	Pre-2023	—	—	—	—	—	0.318	0.617
XGBoost	Pre-2024	—	—	—	—	—	—	**0.585**
Logistic (L2)	Pre-2018	0.688	**0.724**	0.292	0.198	**0.200**	0.332	0.608
Logistic (L2)	Pre-2019	—	**0.686**	0.318	0.200	**0.200**	0.237	0.271
Logistic (L2)	Pre-2020	—	—	0.331	0.175	**0.200**	0.295	0.273
Logistic (L2)	Pre-2021	—	—	—	0.165	**0.200**	0.253	0.243
Logistic (L2)	Pre-2022	—	—	—	—	**0.250**	0.225	0.238
Logistic (L2)	Pre-2023	—	—	—	—	—	0.255	0.217
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.276
Deep (MLP)	Pre-2018	0.693	0.555	0.451	0.132	0.005	0.020	0.034
Deep (MLP)	Pre-2019	—	0.260	0.419	0.111	0.009	0.020	0.041
Deep (MLP)	Pre-2020	—	—	0.465	0.119	0.010	0.020	0.052
Deep (MLP)	Pre-2021	—	—	—	0.122	0.009	0.021	0.070
Deep (MLP)	Pre-2022	—	—	—	—	0.011	0.025	0.223
Deep (MLP)	Pre-2023	—	—	—	—	—	0.246	0.167
Deep (MLP)	Pre-2024	—	—	—	—	—	—	0.253

Table 33: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	**6.412**	7.050	**10.101**	10.554	21.600	31.084	31.418
CatBoost	Pre-2019	—	10.177	**9.834**	10.223	16.615	**31.346**	39.193
CatBoost	Pre-2020	—	—	**9.858**	**10.592**	36.000	**30.150**	32.474
CatBoost	Pre-2021	—	—	—	**13.421**	**43.200**	**36.850**	25.915
CatBoost	Pre-2022	—	—	—	—	43.200	**29.253**	26.491
CatBoost	Pre-2023	—	—	—	—	—	**28.437**	**42.925**
CatBoost	Pre-2024	—	—	—	—	—	—	25.233
XGBoost	Pre-2018	6.099	8.617	9.120	**12.808**	**43.200**	**47.737**	**47.703**
XGBoost	Pre-2019	—	7.913	7.108	**13.876**	19.636	18.071	**51.062**
XGBoost	Pre-2020	—	—	7.570	7.741	24.000	22.194	**49.271**
XGBoost	Pre-2021	—	—	—	12.747	30.857	23.929	**36.505**
XGBoost	Pre-2022	—	—	—	—	**54.000**	20.180	**27.995**
XGBoost	Pre-2023	—	—	—	—	—	15.977	33.172
XGBoost	Pre-2024	—	—	—	—	—	—	**31.418**
Logistic (L2)	Pre-2018	4.840	**11.695**	5.739	7.050	**43.200**	16.707	32.698
Logistic (L2)	Pre-2019	—	**11.075**	6.250	7.147	**43.200**	11.913	14.589
Logistic (L2)	Pre-2020	—	—	6.519	6.235	**43.200**	14.814	14.671
Logistic (L2)	Pre-2021	—	—	—	5.884	**43.200**	12.721	13.078
Logistic (L2)	Pre-2022	—	—	—	—	**54.000**	11.287	12.810
Logistic (L2)	Pre-2023	—	—	—	—	—	12.812	11.646
Logistic (L2)	Pre-2024	—	—	—	—	—	—	14.840
Deep (MLP)	Pre-2018	4.873	8.961	8.873	4.710	1.000	1.000	1.828
Deep (MLP)	Pre-2019	—	4.207	8.235	3.956	2.038	1.000	2.207
Deep (MLP)	Pre-2020	—	—	9.152	4.228	2.057	0.986	2.813
Deep (MLP)	Pre-2021	—	—	—	4.364	2.038	1.057	3.787
Deep (MLP)	Pre-2022	—	—	—	—	2.374	1.263	11.963
Deep (MLP)	Pre-2023	—	—	—	—	—	12.378	8.966
Deep (MLP)	Pre-2024	—	—	—	—	—	—	13.583

Table 34: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	2018	2019	2020	2021	2022	2023	2024
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.911 (Tree)	0.724 (Regularized Linear)	0.514 (Tree)	0.359 (Tree)	0.200 (Regularized Linear)	0.950 (Tree)	0.887 (Tree)
Pre-2019	—	0.686 (Regularized Linear)	0.500 (Tree)	0.389 (Tree)	0.200 (Regularized Linear)	0.624 (Tree)	0.950 (Tree)
Pre-2020	—	—	0.501 (Tree)	0.297 (Tree)	0.200 (Regularized Linear)	0.600 (Tree)	0.917 (Tree)
Pre-2021	—	—	—	0.376 (Tree)	0.200 (Regularized Linear)	0.733 (Tree)	0.679 (Tree)
Pre-2022	—	—	—	—	0.250 (Regularized Linear)	0.582 (Tree)	0.521 (Tree)
Pre-2023	—	—	—	—	—	0.566 (Tree)	0.799 (Tree)
Pre-2024	—	—	—	—	—	—	0.585 (Tree)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.41 (Tree)	11.70 (Regularized Linear)	10.10 (Tree)	12.81 (Tree)	43.20 (Regularized Linear)	47.74 (Tree)	47.70 (Tree)
Pre-2019	—	11.07 (Regularized Linear)	9.83 (Tree)	13.88 (Tree)	43.20 (Regularized Linear)	31.35 (Tree)	51.06 (Tree)
Pre-2020	—	—	9.86 (Tree)	10.59 (Tree)	43.20 (Regularized Linear)	30.15 (Tree)	49.27 (Tree)
Pre-2021	—	—	—	13.42 (Tree)	43.20 (Regularized Linear)	36.85 (Tree)	36.51 (Tree)
Pre-2022	—	—	—	—	54.00 (Regularized Linear)	29.25 (Tree)	27.99 (Tree)
Pre-2023	—	—	—	—	—	28.44 (Tree)	42.93 (Tree)
Pre-2024	—	—	—	—	—	—	31.42 (Tree)

A.4 Feature Contribution Exhibit

This appendix reports the case-level feature contribution decomposition for the primary filing-date model as a descriptive model-reliance exhibit only. It is included to show the within-model distribution of signed feature contributions, not to support causal interpretation or a portable substantive narrative.

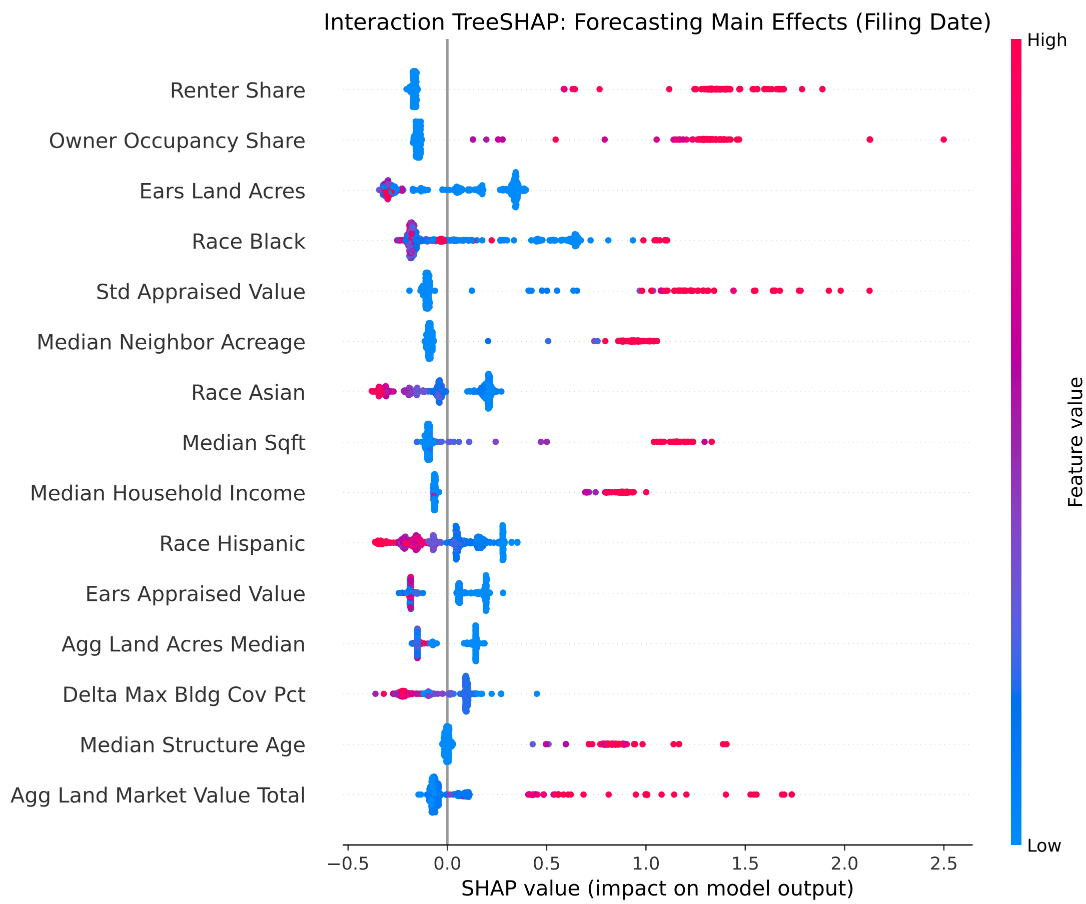

Figure 14: Feature contribution decomposition for the filing-date opposition model using Interaction TreeSHAP (main effects). Each point is a case-level contribution for one predictor; horizontal position gives the signed effect on the model score and vertical density shows the distribution of contributions across cases. Main effects are extracted from the interaction value diagonal, separating each feature's direct contribution from contingent interaction effects with other predictors. Because correlated predictors can exchange attribution weight, this exhibit is descriptive support for within-model behavior rather than a stand-alone substantive ranking of stable feature importance.

Figure 15: Pairwise feature interaction structure for the filing-date opposition model. Each cell (i, j) plots the case-level Interaction TreeSHAP value distribution for the feature pair. Vertical position shows the interaction effect magnitude and direction on the model score, while color indicates the conditioning value of feature j (red for high, blue for low). This reveals not just the strength, but the precise structural form of contingent effects (e.g., whether high values of feature j amplify or suppress the effect of feature i). This exhibit complements Figure 14 by revealing the collinearity and interaction structure that the main-effects decomposition controls for.

A.5 Causal Forest Treatment Effect Attribution

This appendix reports Interaction TreeSHAP exhibits for the `CausalForestDML` model, computed via direct access to the internal `econml.grf.CausalForest` estimator. SHAP interaction values are computed exactly using TreeSHAP ($N = 200$ survivor cases). The beeswarm shows main effects on the treatment effect surface (heterogeneity in how features modulate the height attrition penalty); the beeswarm interaction grid reveals the structural form of contingent effects between feature pairs on the treatment surface.

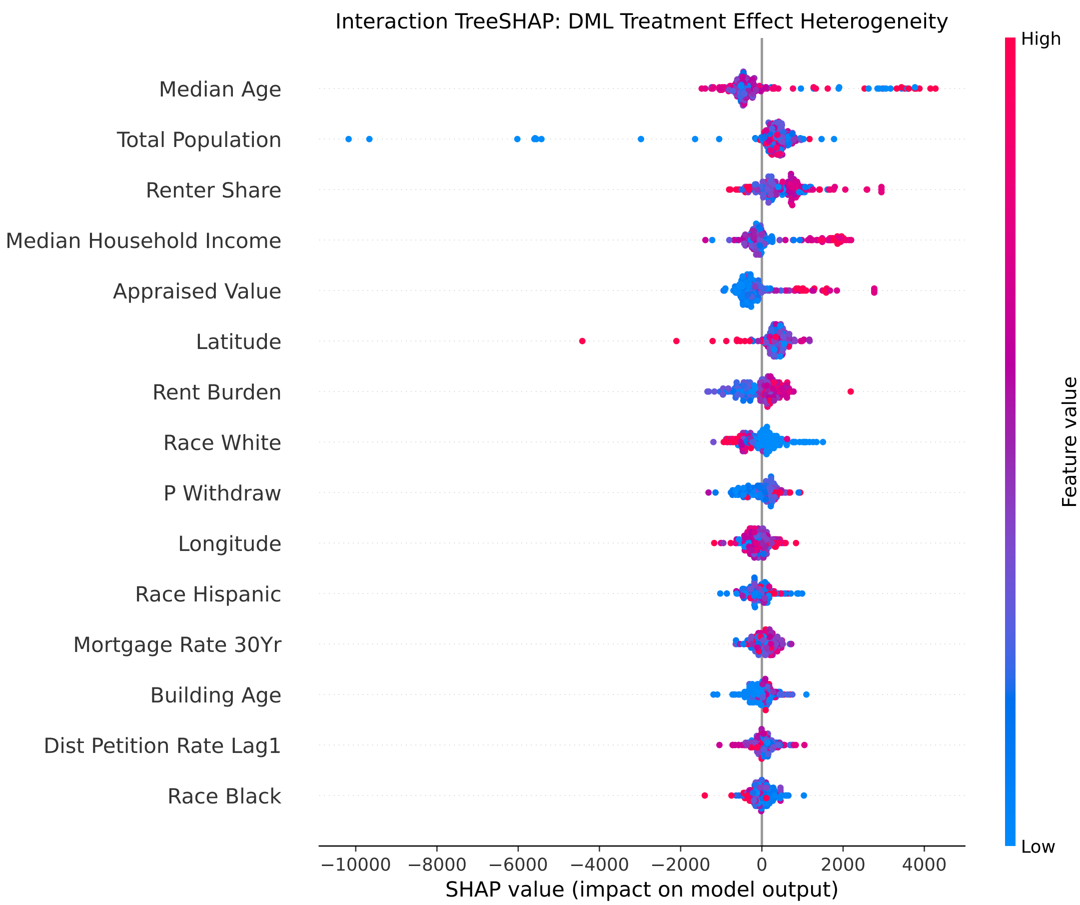

Figure 16: Treatment effect heterogeneity for the Causal Forest DML model, decomposed via Interaction TreeSHAP main effects. Each point represents one case; horizontal position shows the feature's contribution to the local treatment effect estimate. Features with wide distributions indicate strong heterogeneity in how the petition dose penalty operates across the sample.

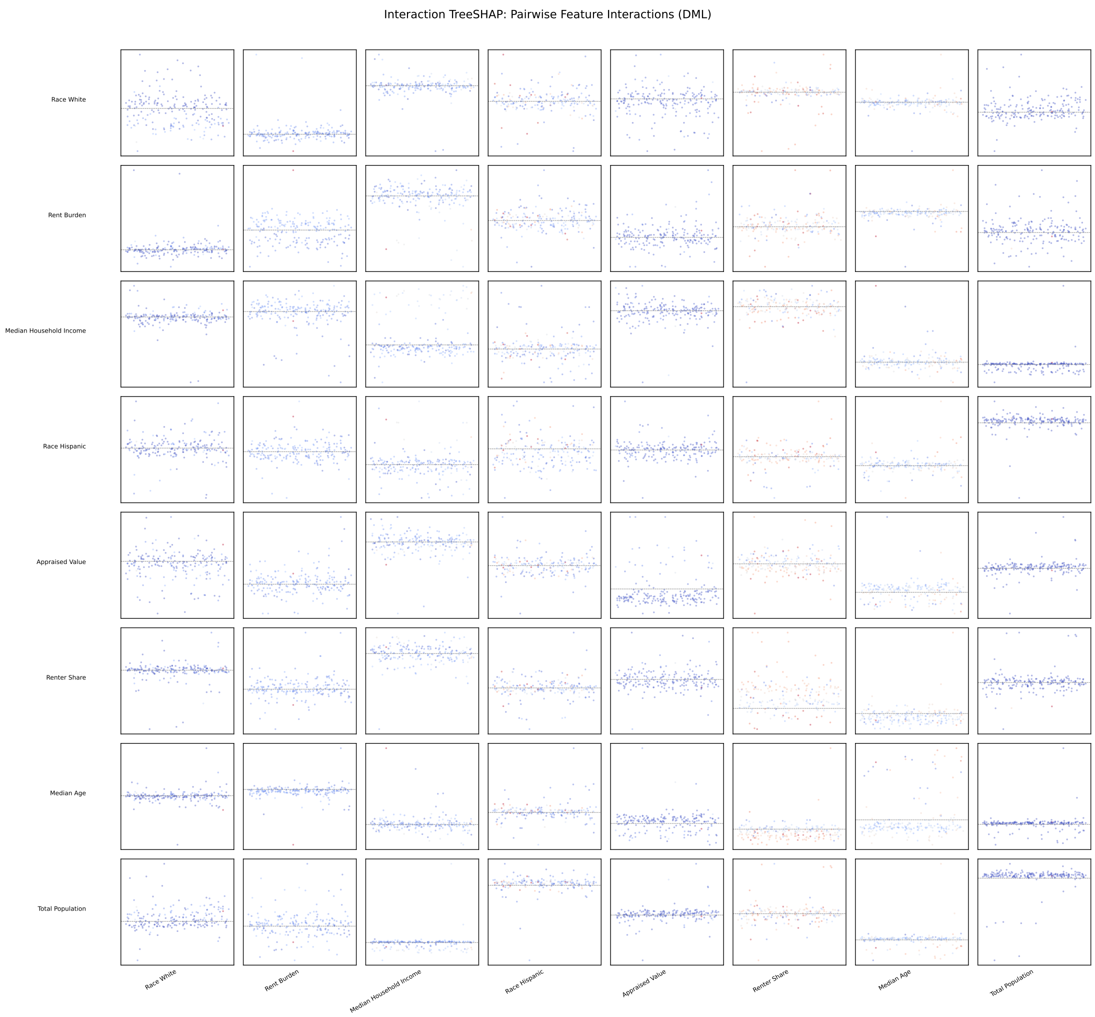

Figure 17: Pairwise Interaction TreeSHAP structure for the Causal Forest DML treatment effect surface. Each cell (i, j) plots the case-level interaction value distribution across $N = 200$ cases, showing how the treatment effect heterogeneity attributable to feature i is conditioned by the realized value of feature j (colored red for high, blue for low). This structural view reveals which feature pairs jointly modulate the petition dose penalty.

A.6 Referee Robustness Checks

Following the structured sensitivity framework recommended for causal machine learning submissions (Chernozhukov et al., 2018), we report four orthogonal robustness checks targeting the failure modes most likely to generate referee skepticism: nuisance model dependence, seed sensitivity, treatment definition sensitivity, and covariate group omission. All results are generated via our evaluation pipeline against the hardened DML panel described in Section 8.4.

A.6.1 Nuisance Model Sensitivity

The DML estimator is sensitive to the quality of nuisance learners ($\hat{m}(X)$ and $\hat{g}(X)$). Table 35 reports the ATE on height attrition across three learner families: linear (Lasso), Random Forest, and Gradient Boosting. Sign consistency across non-parametric learners and magnitude divergence from the linear specification reflect the non-linearity identified by the primary Causal Forest, supporting the specification choice.

Table 35: Height CATE across nuisance learner families. DML with honest cross-fitting; continuous petition dose treatment.

Nuisance Learner	**Height CATE (feet)**
Linear / Lasso	81.22
Random Forest	1.17
Gradient Boosting	-9.65

The Lasso divergence from tree-based learners is consistent with strong non-linearity in the outcome function. Causal Forest (primary) is a non-parametric nuisance estimator and is the preferred specification.

A.6.2 Multi-Seed Stability (N = 20)

To rule out lucky-draw estimation, the Causal Forest is re-estimated across 20 independent random seeds with a fixed data and pipeline specification. The mean ATE, standard deviation, and sign consistency rate are reported in Table 36.

The wide standard deviation (45.14 feet) and near-symmetric sign consistency (45% positive) of the *aggregate* ATE are expected and do not indicate estimator failure. When true causal effects are genuinely heterogeneous – as documented in Section 8.10, where $\hat{\tau}(x_i)$ ranges from -4.72 to 35.69 feet across the parcel distribution – a single aggregate CATE is poorly identified even with a fixed specification, because marginal sampling variation in which cases are included shifts the aggregate substantially. The Causal Forest is designed to estimate the *function* $\hat{\tau}(x_i)$, not a scalar point treatment effect; the correct summary is the CATE quartile distribution in Table 13, where the median is stably 3.35 feet and 91.8% of parcels show a consistent sign. The aggregate CATE instability is itself evidence that the true effect function has high variance across the covariate space.

Table 36: Height CATE distribution across 20 independent random seeds. Wide std dev reflects genuine effect heterogeneity, not estimator instability.

Statistic	**Height CATE (feet)**	**Sign Consistency**
Mean (N=20)	−2.86	45% positive
Std Dev	45.14	n/a

Fixed pipeline specification. Only random seed varies. See Section 8.10 for the CATE distribution.

A.6.3 Treatment Definition Robustness

To assess whether results are an artifact of treatment scaling, the ATE is estimated under four alternative treatment definitions: continuous petition dose (baseline), log-transformed dose, winsorized dose (95th percentile cap), and binary threshold ($T > 0.20$). Sign and magnitude stability across specifications is reported in Table 37.

Table 37: Height CATE under alternative treatment specifications.

Treatment Specification	**Height CATE**
Continuous (Baseline)	81.22
Log-Transformed	89.76
Winsorized (95th pctile)	N/A
Binary (> 0.20)	−32.11

A.6.4 Feature Ablation Study

To assess whether the effect is driven by any single covariate group, we sequentially remove four feature groups from X_i and re-estimate. Table 38 reports the ATE under each ablation, with the baseline full-X estimate as a reference.

Table 38: Height CATE under successive pre-treatment covariate group removals.

Specification	**Height CATE**
Baseline (Full X)	81.22
No Spatial Features	110.87
No Macro Variables	97.07
No History / Lags	101.34
No Demographics	119.19

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